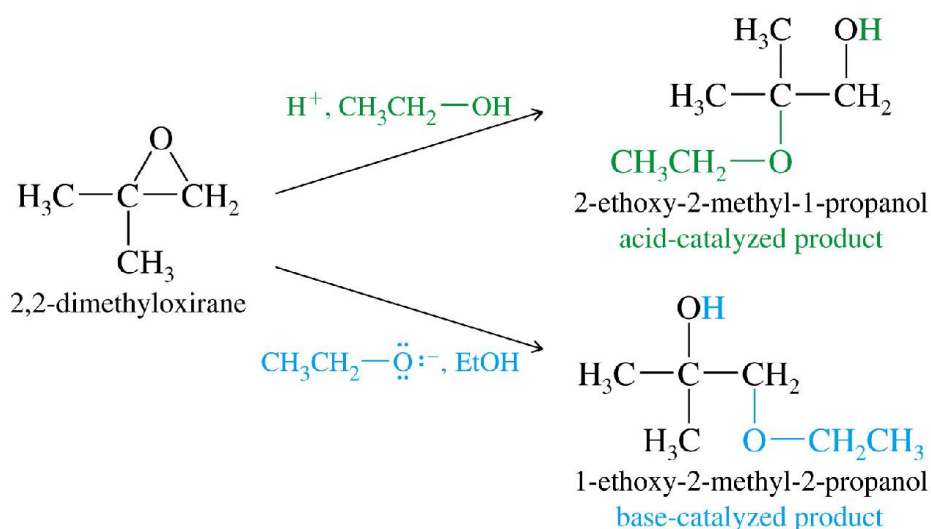


Table of Contents

ALCOHOLS, ETHERS AND PHENOLS

↪	Theory	2
↪	Exercise - 1 : Basic Objective Questions	18
↪	Exercise - 2 : Previous Year JEE Mains Questions	25
↪	Exercise - 3 : Advanced Objective Questions	29
↪	Exercise - 4 : Previous Year JEE Advanced Questions	37
↪	Answer Key	44



THEORY

1. INTRODUCTION

The compounds in which hydroxyl group ($-\text{OH}$) is attached to a saturated carbon atom are called as **Alcohols**. The compounds in which a hydroxyl group is attached to an unsaturated carbon atom of a double bond are called as **Enols**. The saturated carbon may be that of an alkyl, alkenyl, alkynyl, cycloalkyl or benzyl group. However, if a hydroxyl group is attached to a benzene ring, the compounds are called as **Phenols**.

The alcohols are further classified as : **Monohydric** (containing one $-\text{OH}$ group), **Dihydric** (containing two $-\text{OH}$ groups) and **Trihydric** (containing three $-\text{OH}$ groups).

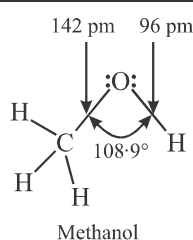
Alcohols find usage in industry as well as day to day life. For example, ordinary spirit used for polishing wooden furniture is chiefly ethanol. Sugar, cotton, paper are all made up of compounds containing $-\text{OH}$ groups. Phenols are used in several important polymers such as *Bakelite* and in several drugs such as *Aspirin*. Ethers are commonly used as solvents and anaesthetics.

2. CLASSIFICATION

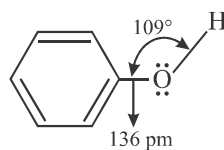
CH_3-OH	Methanol	
$\text{R}-\text{CH}_2-\text{OH}$	1° Alcohol	Primary
$\begin{array}{c} \text{R} \\ \\ \text{R}-\text{CH}-\text{OH} \end{array}$	2° Alcohol	Secondary
$\begin{array}{c} \text{R} \\ \\ \text{R}-\text{C}-\text{OH} \\ \\ \text{R} \end{array}$	3° Alcohol	Tertiary
$\text{R}-\text{O}-\text{R}$	Symmetrical Ether	
$\text{R}-\text{O}-\text{R}'$	Unsymmetrical Ether	

3. STRUCTURES OF FUNCTIONAL GROUPS

In alcohols, the oxygen of the $-\text{OH}$ group is attached to carbon by a sigma (σ) bond formed by the overlap of a sp^3 hybridised orbital of carbon with a sp^3 hybridised orbital of oxygen. The following figure depicts structural aspects of methanol, phenol and methoxymethane.

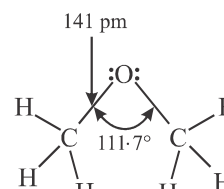


Bond angle is slightly less due to LP-LP repulsion.



Phenol

Lone pair of oxygen is delocalised on ring due to which $\text{C}-\text{O}$ bond length is less.



Methoxymethane

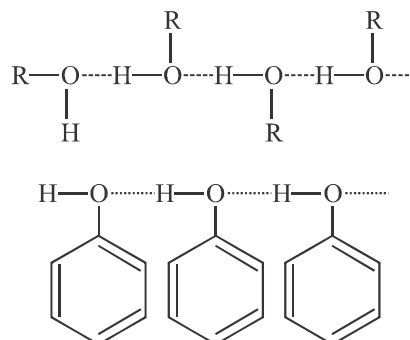
Bond angle in methoxymethane is more than tetrahedral angle due to repulsive interaction between the two bulky (R) groups. The $\text{C}-\text{O}$ bond length is same as in alcohols.

4. PHYSICAL PROPERTIES

4.1 Boiling Point

The boiling points of alcohols and phenols increase with increase in the number of carbon atoms (increase in van der Waals forces). In alcohols, the boiling point decreases with increase in branching (decrease in Van der Waals forces due to decrease in surface area).

The $-\text{OH}$ group in alcohols and phenols contains a hydrogen, bonded to an electronegative oxygen atom. Therefore, it is capable of forming intermolecular hydrogen bond, strength of which is even greater than amine.



It is due to the presence of strong intermolecular hydrogen bonding that alcohols and phenols have higher boiling points corresponding to other classes of compounds, namely, hydrocarbons, ethers and haloalkanes/haloarenes, amines of comparable molecular masses.

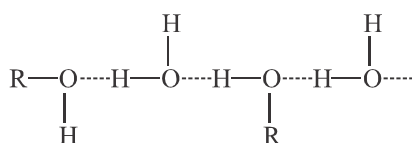
Their boiling points are lower than carboxylic acid which have even more strong H-bond. For isomeric alcohols boiling points decreases with increase in branching due to decrease in van der Waals forces with decrease in size. **The order of boiling point is 1° alcohol $>$ 2° alcohol $>$ 3° alcohol.**

In ethers, boiling point is very less, and comparable to those of alkane of comparable molecular mass due to less dipole moment and absence of H-bonding.

4.2 Solubility

The solubility of alcohols and phenols in water is due to their ability to form hydrogen bonds with water molecules. The solubility decreases with increase in the size of hydrophobic group (R). Higher alcohol are insoluble. Branching increases solubility, due to decrease in surface area of non-polar hydrocarbon part.

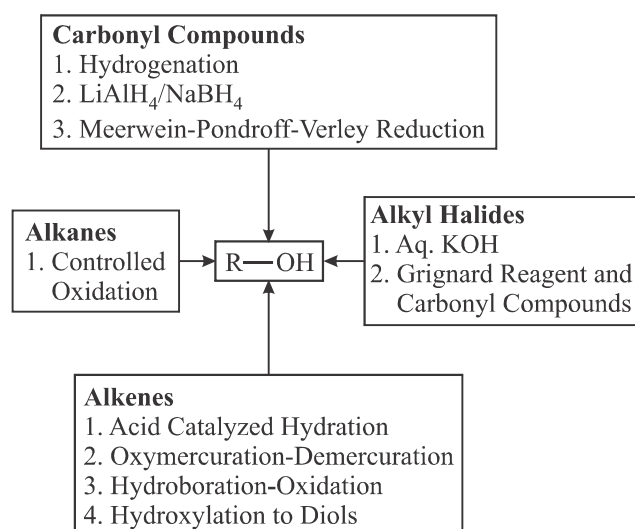
n-butyl alcohol < isobutyl alcohol < sec-butyl alcohol < tert-butyl alcohol



Lower ethers are water soluble but solubility is less than alcohol due to less H-bonding with water and low polarity.

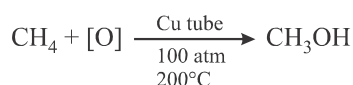
Refer to Table 1, 2, 3 and 4 for reference to physical properties of alcohols, phenols and ethers.

5. PREPARATION OF ALCOHOLS



5.1 Alkane

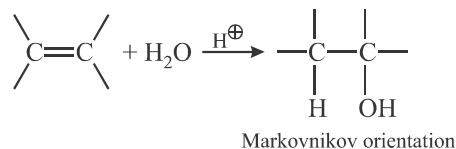
Controlled Oxidation



5.2 Alkenes

5.2.1 Acid Catalyzed Hydration

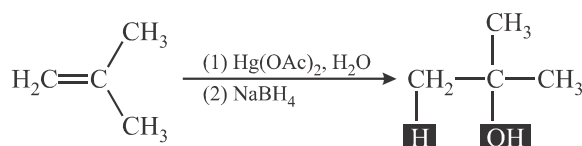
Markovnikov addition with carbocation rearrangements.



5.2.2 Oxymercuration-Demercuration

Markovnikov addition without carbocation rearrangements.

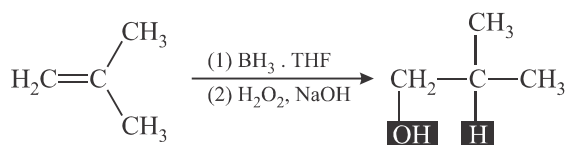
Example - 1



5.2.3 Hydroboration-Oxidation

Anti-Markovnikov addition.

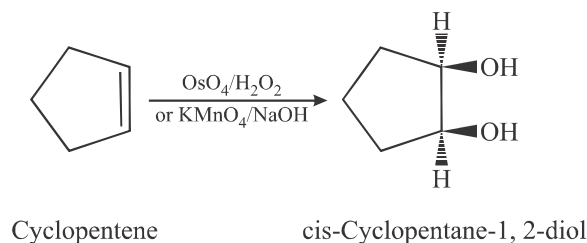
Example - 2



5.2.4 SYN Hydroxylation

Reagents : Cold dil. $\text{KMnO}_4/\text{NaOH}$ or $\text{OsO}_4/\text{H}_2\text{O}_2$

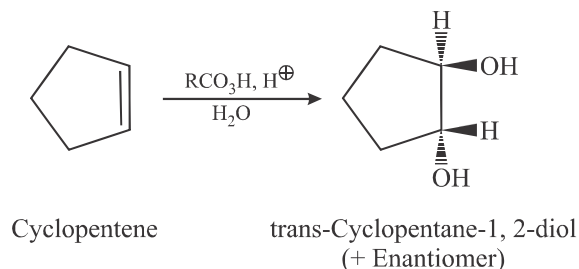
Example - 3



5.2.5 ANTI Hydroxylation

Reagents : Peroxy Acids followed by Acidic Hydrolysis

Example - 4

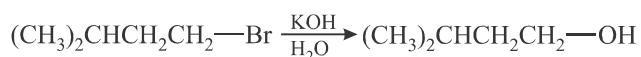


5.3 Alkyl Halide

5.3.1 Second-order Substitution

Primary (and some Secondary) Halides

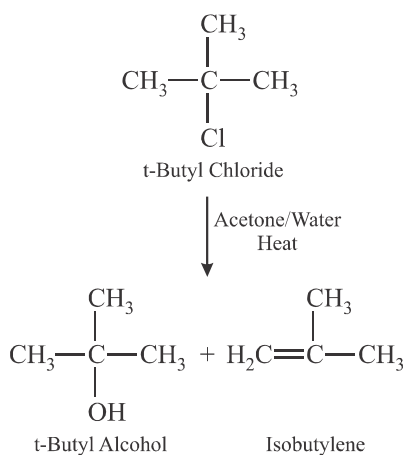
Example - 5



5.3.2 First-order Substitution

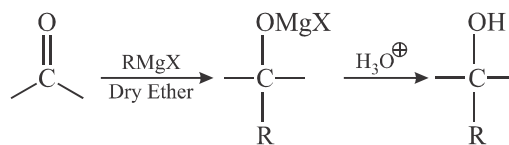
Tertiary (and some Secondary) Halides

Example - 6



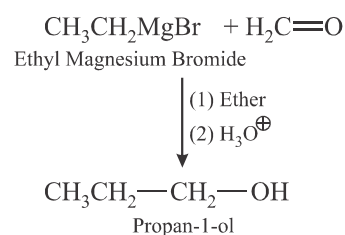
5.3.3 Grignard Reagent/Organolithium Reagent

Nucleophilic addition to the carbonyl group



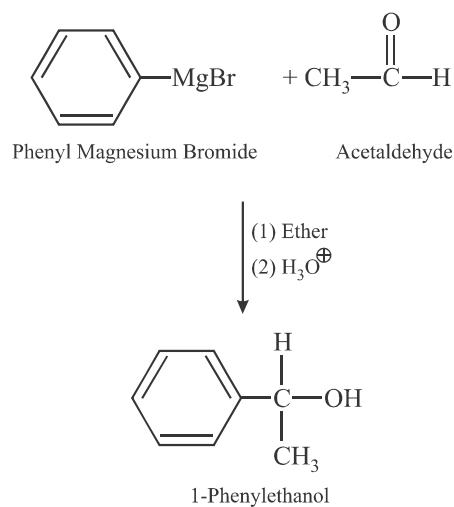
(a) Addition to Formaldehyde - 1° Alcohol

Example - 7



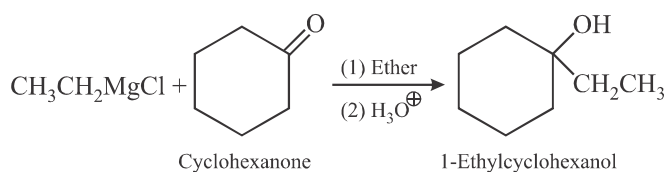
(b) Addition to an Aldehyde - 2° Alcohol

Example - 8



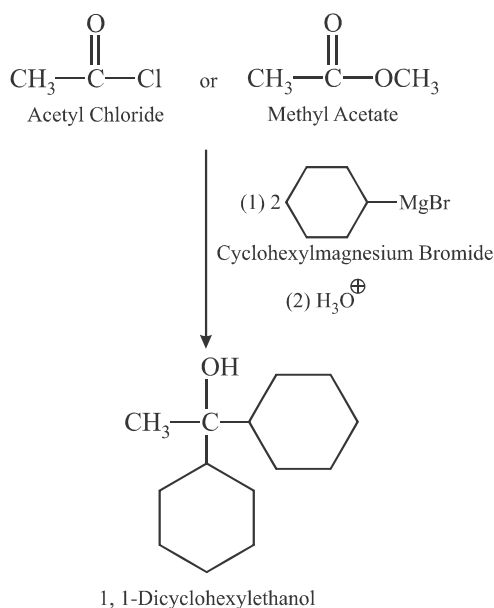
(c) Addition to a Ketone - 3° Alcohol

Example - 9



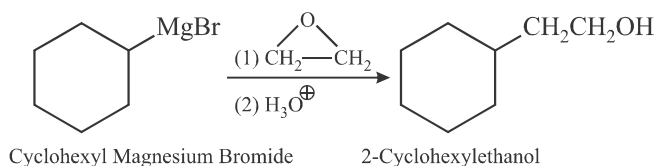
(d) Addition to an Acid Halide or an Ester - 3° Alcohol

Example - 10



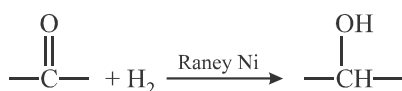
(e) Addition to Ethylene Oxide - 1° Alcohol (with two carbon atoms added)

Example - 11



5.4 Carbonyl Compounds

5.4.1 Catalytic Hydrogenation

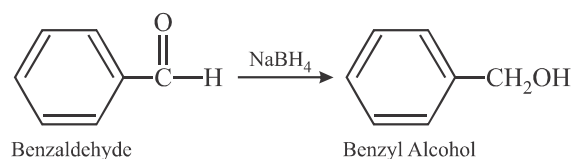


This method is usually not as selective or as effective as the use of hydride reagents.

5.4.2 Reduction with Metal Hydrides

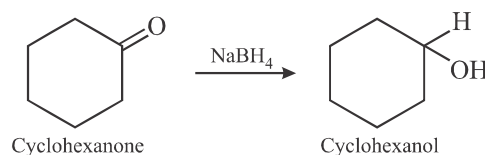
1. Reduction of an aldehyde gives a primary alcohol

Example - 12



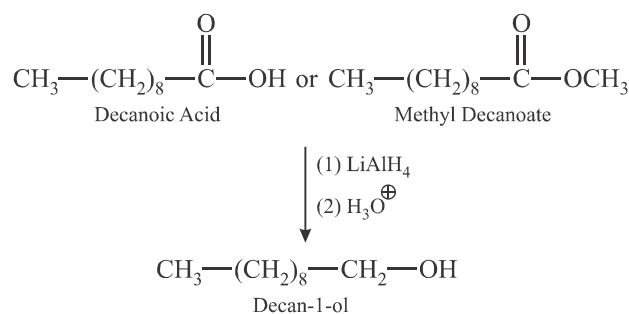
2. Reduction of a ketone gives a secondary alcohol

Example - 13



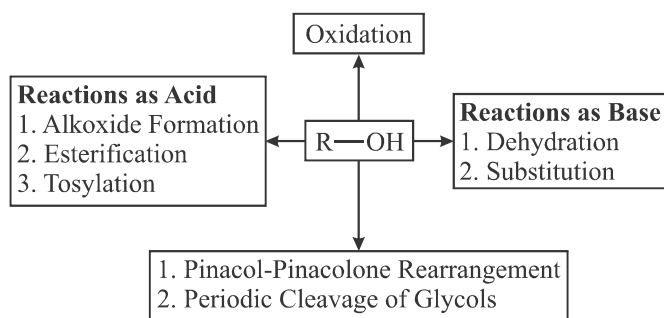
3. Reduction of an acid or ester gives a primary alcohol

Example - 14

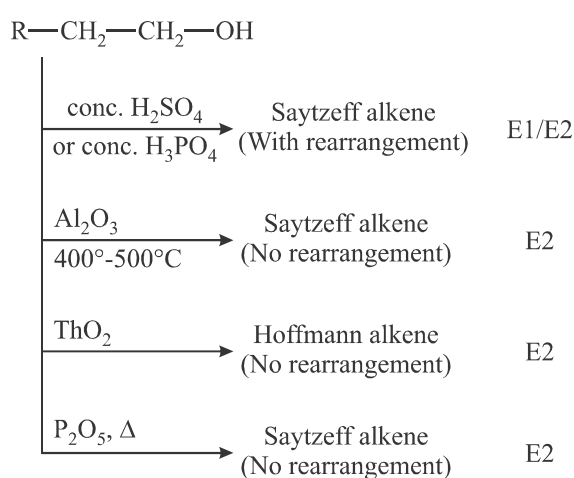

Reactions of LiAlH_4 and NaBH_4

Functional Group	Structure	NaBH_4	LiAlH_4
Aldehyde	$\text{R}-\text{C}(=\text{O})-\text{H}$	$\text{R}-\text{CH}_2-\text{OH}$	$\text{R}-\text{CH}_2-\text{OH}$
Ketone	$\text{R}-\text{C}(=\text{O})-\text{R}'$	$\text{R}-\text{CH}(\text{OH})-\text{R}'$	$\text{R}-\text{CH}(\text{OH})-\text{R}'$
Alkene	$\text{C}=\text{C}$	No Reaction	No Reaction
Acid Anion	$\text{R}-\text{C}(=\text{O})-\text{O}^{\ominus}$ anion in base	No Reaction	$\text{R}-\text{CH}_2-\text{OH}$
Ester	$\text{R}-\text{C}(=\text{O})-\text{OR}'$	No Reaction	$\text{R}-\text{CH}_2\text{OH} + \text{R}'\text{OH}$

6. REACTIONS OF ALCOHOLS

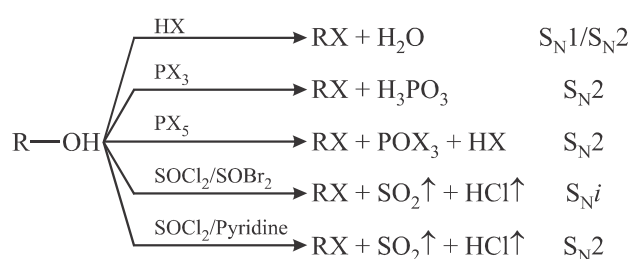


6.1 Dehydration



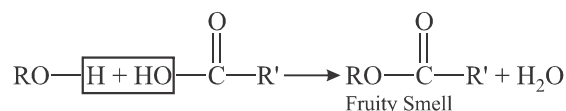
Note... Reactions with Al_2O_3 , ThO_2 and P_2O_5 occur in gaseous phase.

6.2 Substitution



Note... Reaction of alcohol with SOCl_2 in the presence of Pyridine is known as **Darzen's Process**.

6.3 Esterification

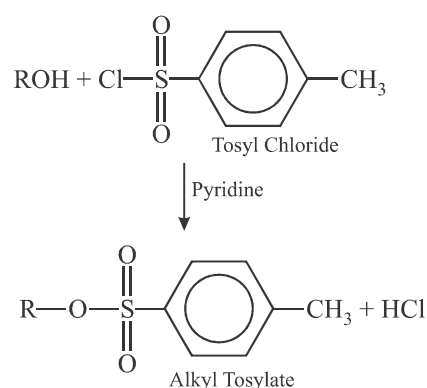


Esterification is catalyzed by an acid or base.

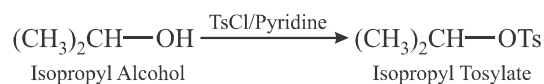
6.4 Tosylation

Tosylation is used to convert poor leaving group OH into good leaving group OTs.

Example - 15

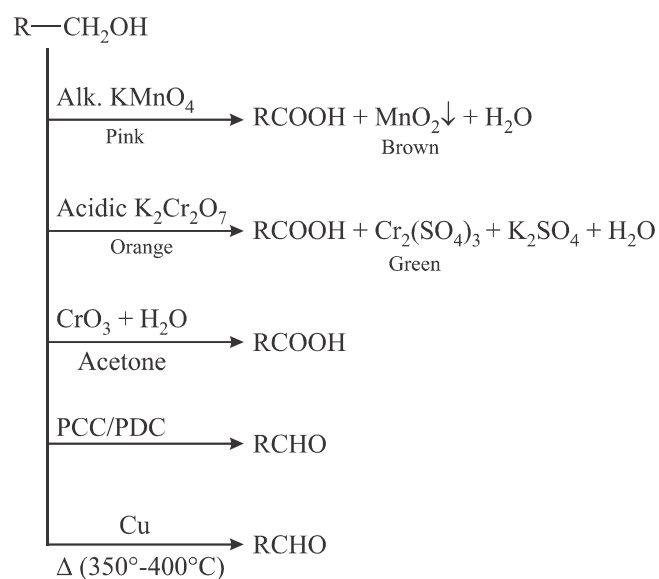


Example - 16



6.5 Oxidation

(a) Primary Alcohols

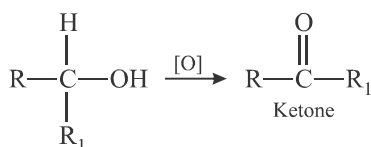




Oxidation of Alcohols : Points to Remember

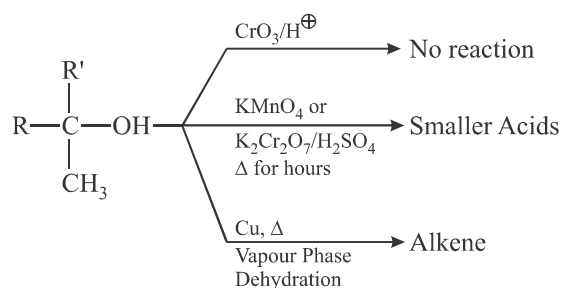
1. Jones Reagent – $\text{CrO}_3 + \text{H}_2\text{O}/\text{Acetone}$
2. PCC – Pyridinium Chlorochromate
3. PDC – Pyridinium Dichromate
4. $\text{CrO}_3/\text{Pyridine}$ in CH_2Cl_2 – Collins Reagent.
5. Cr (VI) oxidizes primary alcohol to aldehyde in anhydrous/ non-aqueous state and to carboxylic acid otherwise.

(b) Secondary Alcohols

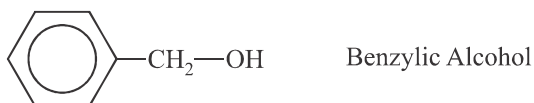
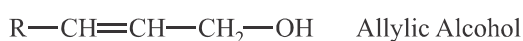


Secondary alcohol in the presence of any oxidising agent is oxidised to ketone.

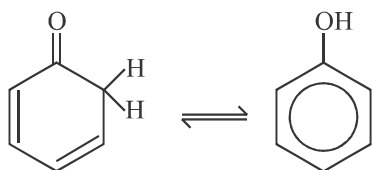
(c) Tertiary Alcohols



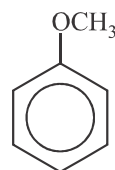
MnO_2 is an oxidising agent which oxidises only allylic, benzylic & propargylic alcohols.



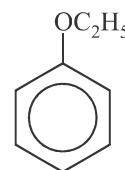
7. PHENOL



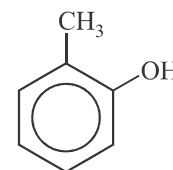
Common Phenols and Aromatic Ethers



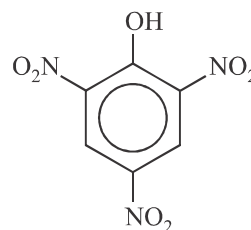
Anisole



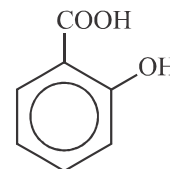
Phenetole



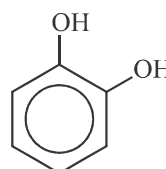
o-Cresol



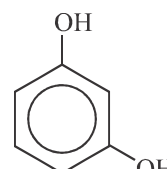
Picric Acid



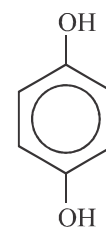
Salicylic Acid



Catechol

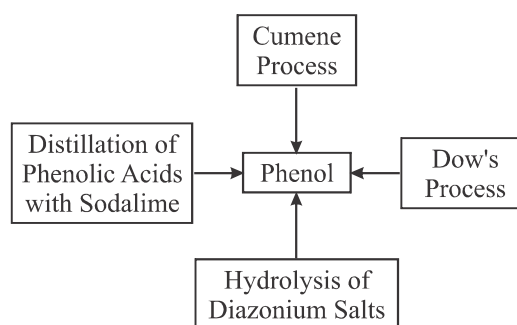


Resocinol



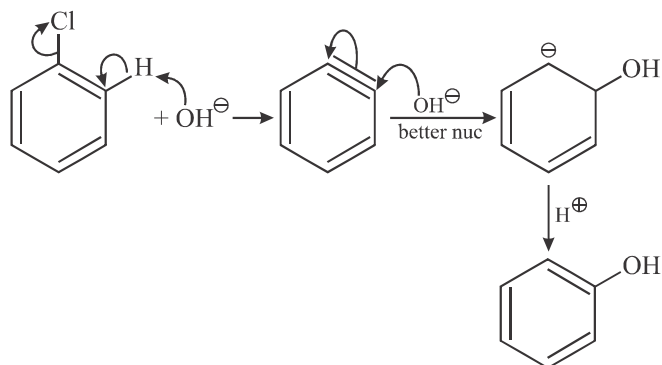
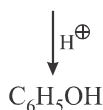
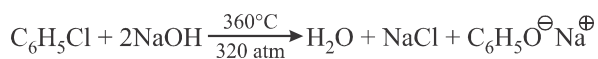
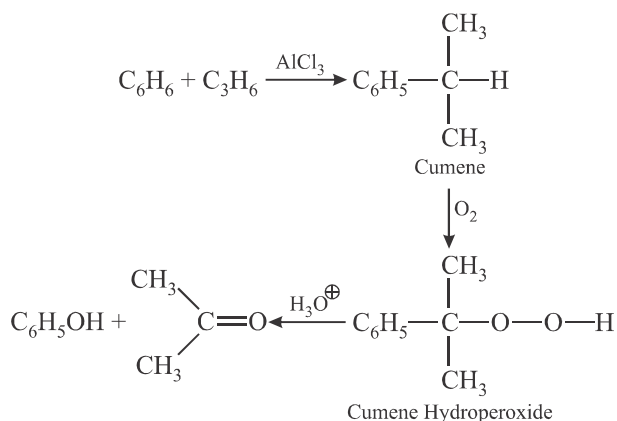
Hydroquinone

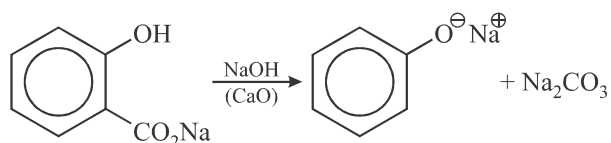
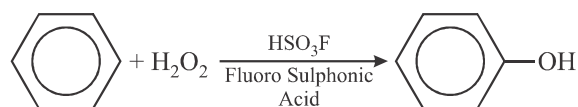
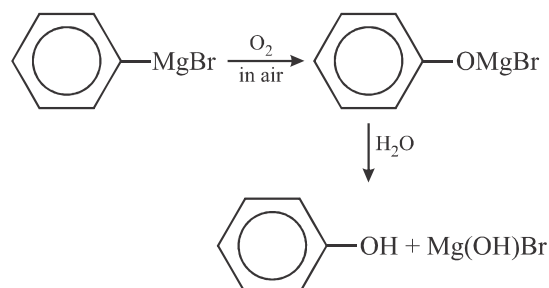
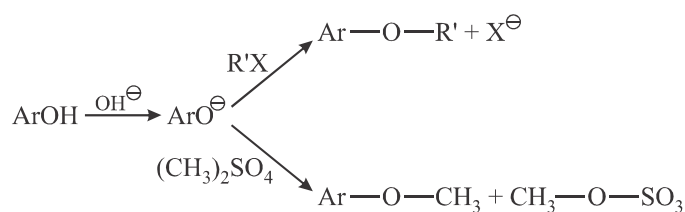
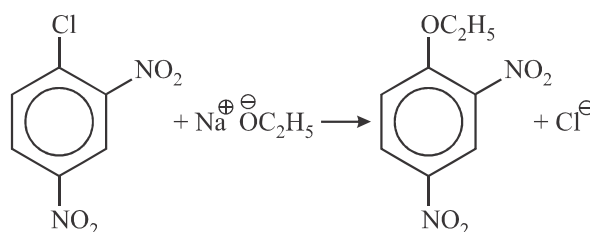
8. PREPARATION OF PHENOLS



8.1 Dow's Process

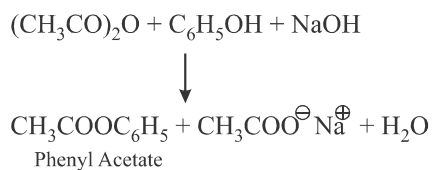
This is industrial method for preparation of phenol. It occurs through Benzyne Mechanism.

Example - 17

8.2 Cumene Process
Example - 18

8.3 Hydrolysis of Diazonium Salts

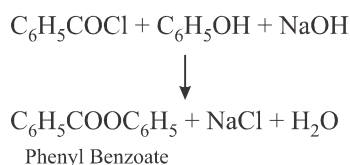
8.4 Distillation of Phenolic Acids with Sodalime
Example - 19

8.5 Benzene
Example - 20

8.6 Grignard Reagent
Example - 21

9. REACTIONS OF PHENOLS
9.1 Formation of Ethers
(a) Williamson Synthesis

(b) Nucleophilic Aromatic Substitution
Example - 22


9.2 Formation of Esters

Example - 23

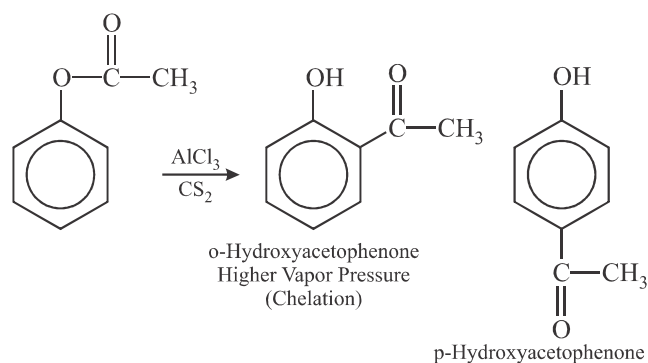


Example - 24



9.3 Fries Rearrangement

Example - 25



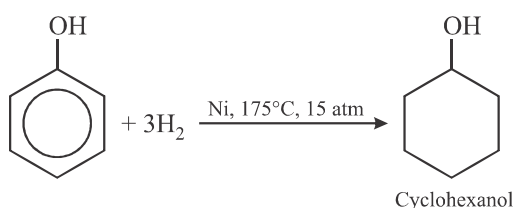
Note... **Kinetic Control versus Thermodynamic Control**

1. Para isomer is the major product at 25°C . It has lower ΔH and is formed more rapidly (rate controlled product).
2. Ortho isomer is the chief product at 165°C (equilibrium controlled product).

9.4 Reactions of Benzene Ring

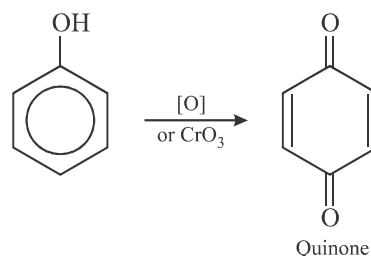
9.4.1 Hydrogenation

Example - 26



9.4.2 Oxidation to Quinones

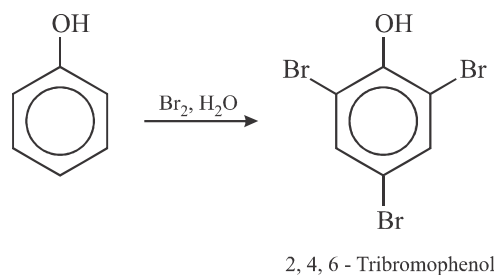
Example - 27



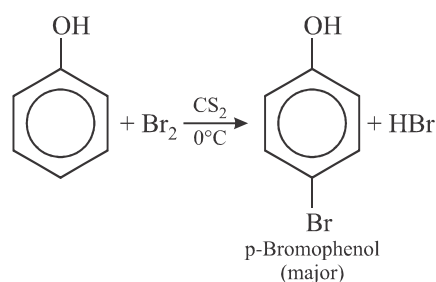
9.4.3 Electrophilic Substitution

(a) Halogenation

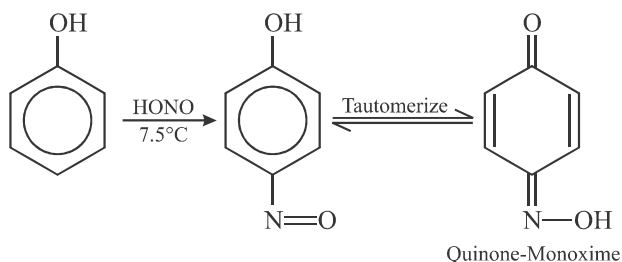
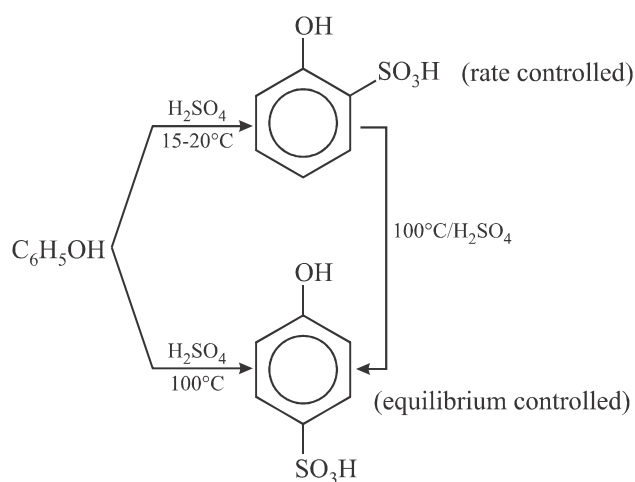
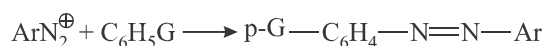
Example - 28



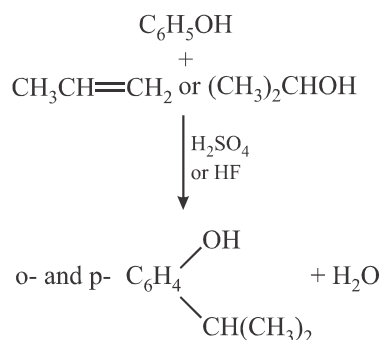
Example - 29



Note... Monobromination is achieved with non-polar solvents such as CS_2 to decrease the electrophilicity of Br_2 and also to minimize phenol ionization. In polar solvents such as water, phenol ionizes to phenoxide ion which is strongly activated and therefore, bromination takes place at all the activated positions.

(b) Nitrosation
Example - 30

(c) Sulphonation
Example - 31

(d) Diazonium Salt Coupling - Azophenols


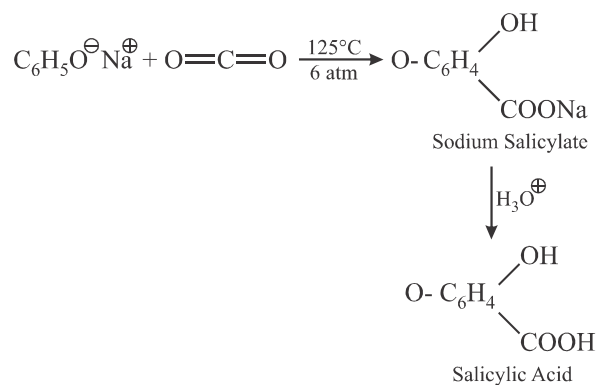
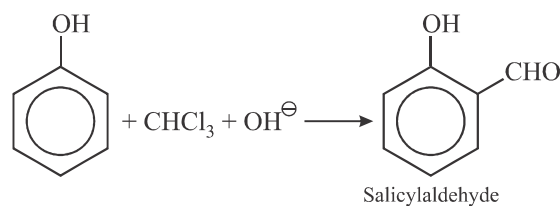
G is an electron releasing groups such as $-\text{OH}$, $-\text{OR}$, $-\text{NR}_2$, $-\text{NHR}$, $-\text{NH}_2$.

(e) Ring Alkylation
Example - 32


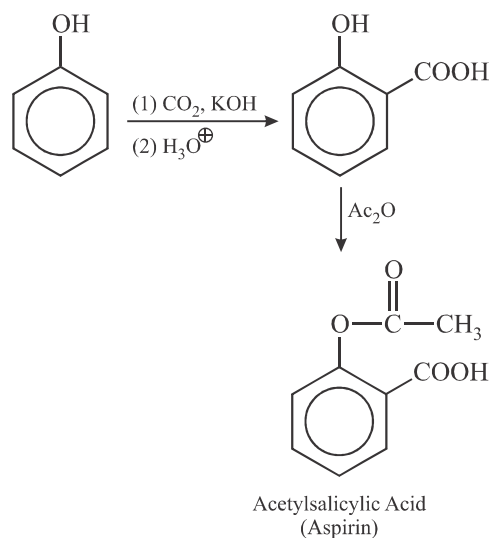
RX and AlCl_3 give poor yields because AlCl_3 co-ordinates with lone pair of oxygen.

(f) Ring Acylation

This is achieved using Fries Rearrangement.

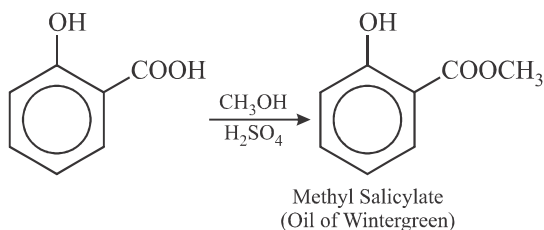
(g) Kolbe's Synthesis
Example - 33

(h) Riemer-Tiemann Synthesis of Phenolic Aldehydes
Example - 34


This reaction involves formation of carbene, CCl_2 .

(i) Formation of Aspirin
Example - 35


(j) Formation of Oil of Wintergreen

Example - 36



10. ETHERS

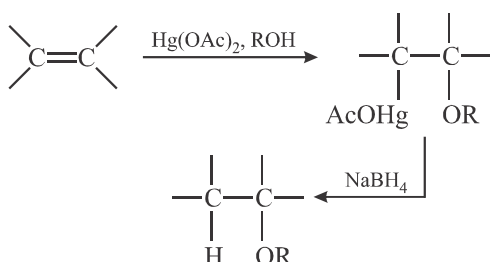
10.1 Williamson Ether Synthesis



Substrate and Leaving Group in Williamson Synthesis

1. Leaving group $X = Cl, Br, I, OTs$, etc.
2. Substrate - Alkyl group R' must be primary.

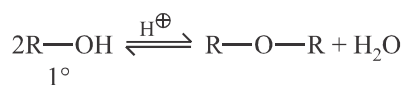
10.2 Alkoxymercuration-Demercuration



This reaction follows Markovnikov orientation.

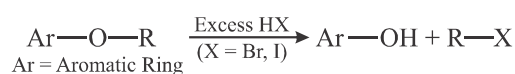
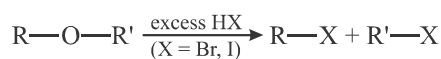
10.3 Bimolecular Dehydration of Alcohols

This reaction is an industrial method for synthesis of ethers.

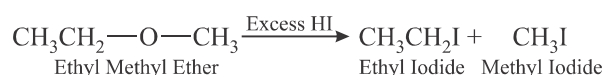


11. REACTIONS OF ETHERS

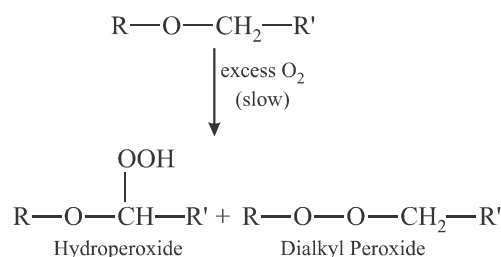
11.1 Cleavage by HBr and HI



Example - 37

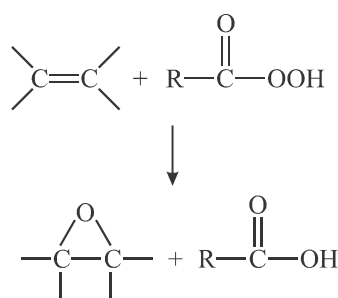


11.2 Autoxidation

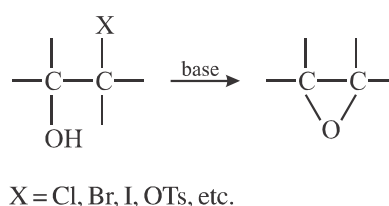


12. PREPARATION OF EPOXIDES

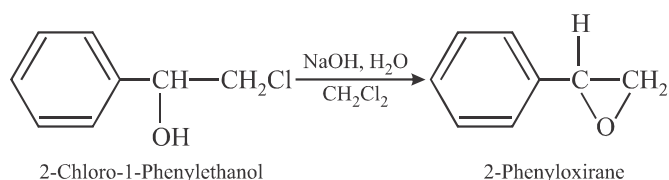
12.1 Peroxy Acid Epoxidation



12.2 Base-promoted cyclization of halohydrins



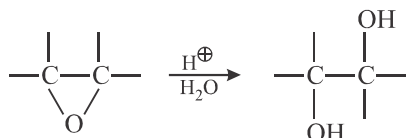
Example - 38



13. REACTIONS OF EPOXIDES

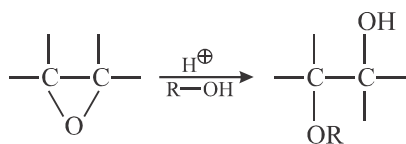
13.1 Acid-catalyzed Opening

(a) In water



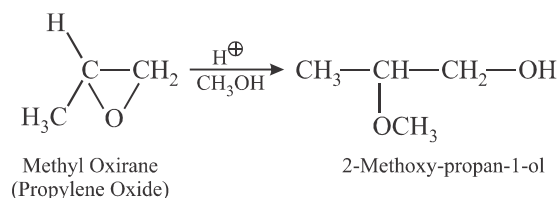
Anti stereochemistry is followed.

(b) In alcohols

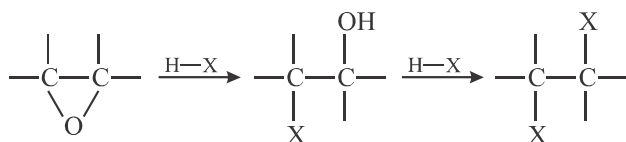


The alkoxy group bonds to the more highly substituted carbon.

Example - 39

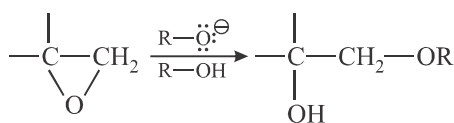


(c) Hydrohalic Acids (X = Cl, Br, I)



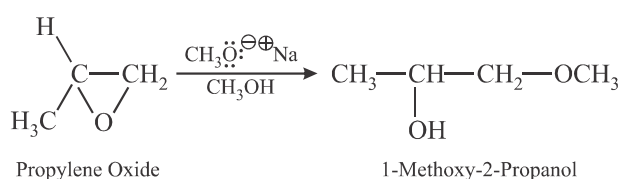
13.2 Base-catalyzed Opening

(a) With Alkoxides

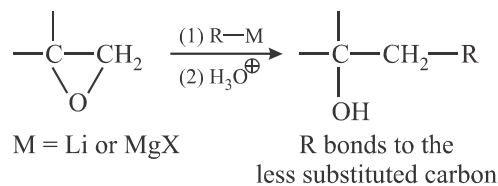


The alkoxy group bonds to the less highly substituted carbon.

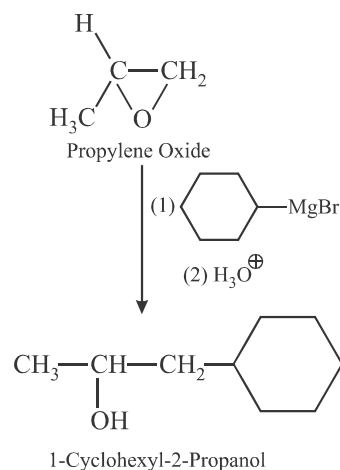
Example - 40



(b) With Organometallics



Example - 41



Opening of Epoxide Ring

1. In acid catalyzed opening, nucleophile attacks on that epoxide carbon from which more stable carbocation can be made.
2. In base catalyzed opening, nucleophile attacks on the less hindered carbon.

14. ACIDIC STRENGTH

(i) Alcohols are weaker acids than thiols even though oxygen is more electronegative than sulphur. Conjugate base of alcohol i.e. RO^- is more basic than RS^- as in RO^- negative charge is placed on smaller oxygen atom so it will have more charge density. But in RS^- negative charge is dispersed on bigger sulphur so it is a poor base and its conjugate acid will be more acidic. Order of acidic strength of some of the compound are given as :



(ii) All alcohols (except CH_3OH) are weaker acids than H_2O due to +I effect of alkyl group. CH_3OH is slightly stronger than H_2O . Phenols are stronger than alcohol due to electron withdrawing benzene ring & resonance stabilized phenoxide ion. Alkoxide ions, the conjugate base of alcohol have no resonance so they are less stable and more basic. Phenol is weaker than carboxylic acid which have strong electron withdrawing carbonyl group and more stable resonating structures.

15. TESTS FOR ALCOHOLS, PHENOLS & ETHERS

15.1 Analysis of Alcohols – Characterization

(a) Alcohols dissolve in cold concentrated sulfuric acid. This property they share with alkenes, amines, practically all compounds containing oxygen, and easily sulfonated compounds. (Alcohols, like other oxygen-containing compounds, form oxonium salts which dissolve in the highly polar sulfuric acid.)

(b) Alcohols are not oxidized by cold dilute, neutral permanganate (although primary and secondary alcohols are oxidized by permanganate under more vigorous conditions.) However, as we have seen, alcohols often contain impurities that are oxidized under these conditions, and so the permanganate test must be interpreted with caution.

(c) Alcohols do not decolorize bromine in carbon tetrachloride. This property serves to distinguish them from alkenes and alkynes.

Alcohols are further distinguished from alkenes and alkynes and from nearly every other kind of compound by their oxidation by chromic anhydride (CrO_3) in aqueous sulfuric acids. Within two seconds, the clear orange solution turns blue-green and becomes opaque.



Tertiary alcohols do not give this test. Aldehydes do, but are easily differentiated in other ways.

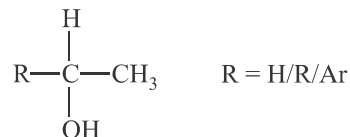
(d) Reactions of alcohols with sodium metal with the evolution of hydrogen gas is of some use in characterization. A wet compound of any kind, of course, will do the same thing, until the water is used up.

(e) The presence of the $-\text{OH}$ group in a molecule is often indicated by the formation of an ester upon treatment with an acid chloride or anhydride. Some esters are sweet-smelling; others are solids and sharp melting points, and can be derivatives in identifications. If the molecular formulas of starting material and product are determined, it is possible to calculate how many $-\text{OH}$ groups are present.

(f) Whether an alcohol is primary, secondary or tertiary is shown by the **Lucas test**, which is based upon the difference in reactivity of the three classes toward hydrogen halides. Alcohols (of not more than six carbons) are soluble in the Lucas reagent, a mixture of concentrated hydrochloric acid and zinc chloride. The corresponding alkyl chlorides are insoluble. Formation of a chloride from an alcohol is indicated by the cloudiness that appears when the chloride separates from the solution. Hence, the time required for cloudiness to appear is a measure of the reactivity of the alcohol.

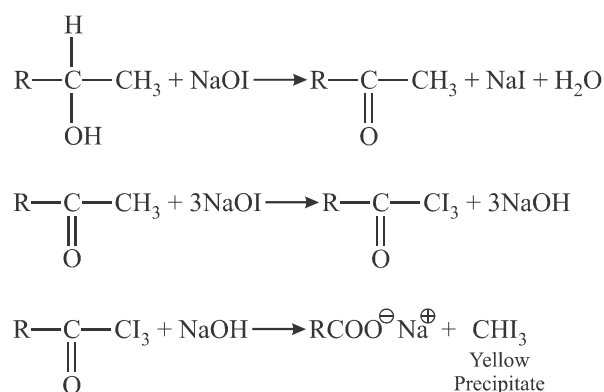
A tertiary alcohol reacts immediately with the Lucas reagent, and a secondary alcohol reacts within five minutes. A primary alcohol does not react appreciably at room temperature. Benzyl alcohol and allyl alcohol react as rapidly as tertiary alcohols with the Lucas reagent. Allyl chloride, however, is soluble in the reagent.

(g) Whether or not an alcohol contains one particular structural unit is shown by the **iodoform test**. The alcohol is treated with iodine and sodium hydroxide (sodium hypoiodite, NaOI). An alcohol of the structure yields a yellow precipitate of iodoform (CHI_3 m.p. 119°).



Positive Iodoform Test	Negative Iodoform Test
$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{H} \\ \\ \text{OH} \end{array}$	Any other primary alcohol
$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$
$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{CH}_2\text{CH}_2\text{CH}_3 \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_2\text{CH}_3 \\ \\ \text{OH} \end{array}$
$\begin{array}{c} \text{H} \\ \\ \text{C}_6\text{H}_5-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$	$\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2\text{OH}$

The reaction involves oxidation, halogenation and cleavage.



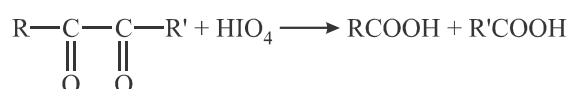
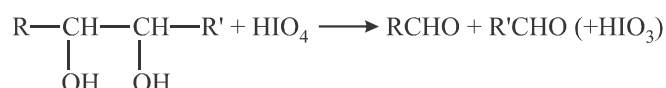
As would be expected from the equations, a compound of structure



In certain special cases this reaction is used not as a test, but to synthesize the carboxylic acid, RCOOH . Here, hypobromite or the cheaper hypochlorite would probably be used.

15.2 Analysis of Glycols, Periodic Acid Oxidation

Upon treatment with periodic acid, HIO_4 , compounds containing two or more $-\text{OH}$ or $\text{C}=\text{O}$ groups attached to adjacent carbon atoms undergo oxidation with cleavage of carbon-carbon bonds.



15.3 Miscellaneous Tests

(a) **Ceric Ammonium Nitrate Test** - Alcohols give red colour with this reagent.



(b) **Potassium Dichromate Test** - Alcohols turn orange dichromate to green. Tertiary Alcohol do not give this test.

(c) **Ester Test** - Alcohol gives fruity smell of ester with carboxylic acid.

(d) Methanol forms oil of winter green with salicylic acid.

15.4 Victor Meyer's Test

1° Alcohol	2° Alcohol	3° Alcohol
RCH_2OH $\downarrow \text{P} + \text{I}_2$ RCH_2I $\downarrow \text{AgNO}_2$ RCH_2NO_2 $\downarrow \text{HONO}$ $\text{R}-\text{C}-\text{NO}_2$ $ $ $\text{N}-\text{OH}$ Nitrolic acid $\downarrow \text{NaOH}$ Blood Red Colour	$\begin{array}{c} \text{R} \\ \\ \text{CHOH} \\ \\ \text{R} \end{array}$ $\downarrow \text{P} + \text{I}_2$ $\begin{array}{c} \text{R} \\ \\ \text{CH-I} \\ \\ \text{R} \end{array}$ $\downarrow \text{AgNO}_2$ $\begin{array}{c} \text{R} \\ \\ \text{CH-NO}_2 \\ \\ \text{R} \end{array}$ $\downarrow \text{HONO}$ $\begin{array}{c} \text{R} \\ \\ \text{C-NO}_2 \\ \\ \text{N=O} \end{array}$ Pseudonitrol $\downarrow \text{NaOH}$ Blue Colour	$\begin{array}{c} \text{R} \\ \\ \text{C-OH} \\ \\ \text{R} \end{array}$ $\downarrow \text{P} + \text{I}_2$ $\text{R}_3\text{C-I}$ $\downarrow \text{AgNO}_2$ $\text{R}_3\text{C-NO}_2$ $\downarrow \text{HONO}$ No Reaction $\downarrow \text{NaOH}$ Colourless

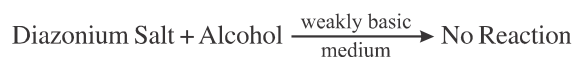
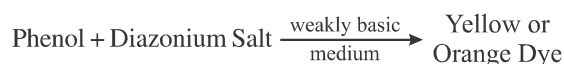
15.5 Differentiation Tests

15.5.1 Alcohols and Phenols

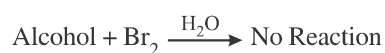
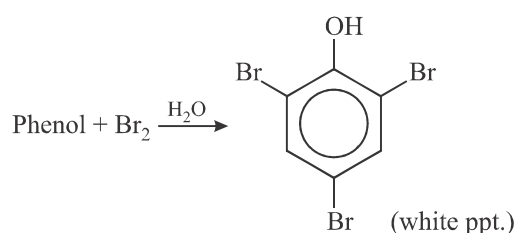
1. **Litmus Test** : Phenol turns blue litmus red but not alcohols.

2. **FeCl_3 Test** : Phenol $\xrightarrow{\text{Neutral FeCl}_3}$ Blue-Violet

3. **Coupling reaction** :



4. **Bromine Water Test** :



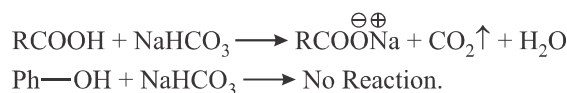
15.5.2 Alcohols and Ethers

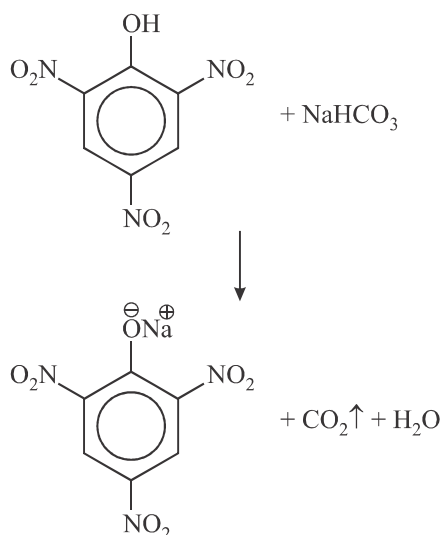
(a) Alcohols react with Na to give H_2 but not ethers.

(b) Alcohols give fumes of HCl with PCl_5 but not ethers.

15.5.3 Sodium Bicarbonate Test

Phenol, ROH and H_2O do not displace CO_2 from carbonate & bicarbonates but RCOOH & RSO_3H gives brisk effervescence of CO_2 which proves that RCOOH & RSO_3H are stronger acids H_2CO_3 but phenol is weaker acid than H_2CO_3 . Nitrophenols also give effervescence of CO_2 with Na_2CO_3 . Trinitrophenol (Picric Acid) is highly acidic due to strong electron withdrawing effect of three groups its acidic strength is comparable to that of carboxylic acids. Its anion is highly resonance stabilised.





15.5.4 FeCl₃ Test

Phenol gives characteristic purple colour with FeCl₃ but alcohols do not react with FeCl₃. Carboxylic acids also form buff coloured precipitate with FeCl₃. Only acetic acid forms red coloured precipitate with FeCl₃, so it can be used as a test for acetate salts.

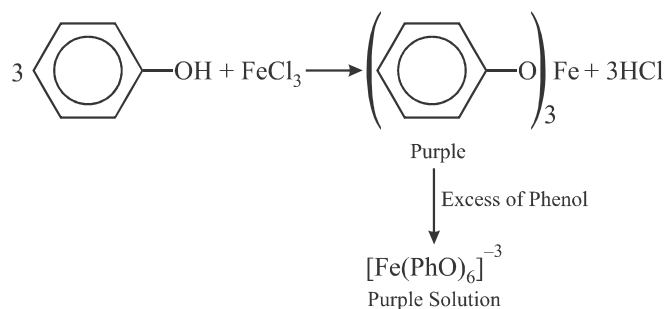


Table 1 : Physical Properties of Selected Alcohols

IUPAC Name	Common Name	Formula	MP(°C)	BP(°C)	Density
methanol	methyl alcohol	CH ₃ OH	- 97	65	0.79
ethanol	ethyl alcohol	CH ₃ CH ₂ OH	- 114	78	0.79
1-propanol	n-propyl alcohol	CH ₃ CH ₂ CH ₂ OH	- 126	97	0.80
2-propanol	isopropyl alcohol	(CH ₃) ₂ CHOH	- 89	82	0.79
1-butanol	n-butyl alcohol	CH ₃ (CH ₂) ₃ OH	- 90	118	0.81
2-butanol	sec-butyl alcohol	CH ₃ CH(OH)CH ₂ CH ₃	- 114	100	0.81
2-methyl-1-propanol	isobutyl alcohol	(CH ₃) ₂ CHCH ₂ OH	- 108	108	0.80
2-methyl-2-propanol	t-butyl alcohol	(CH ₃) ₃ COH	25	83	0.79
1-pentanol	n-pentyl alcohol	CH ₃ (CH ₂) ₄ OH	- 79	138	0.82
3-methyl-1-butanol	isopentyl alcohol	(CH ₃) ₂ CHCH ₂ CH ₂ OH	- 117	132	0.81
2, 2-dimethyl-1-propanol	neopentyl alcohol	(CH ₃) ₃ CCH ₂ OH	52	113	0.81
cyclopentanol	cyclopentyl alcohol	cyclo-C ₅ H ₉ OH	- 19	141	0.95
1-hexanol	n-hexanol	CH ₃ (CH ₂) ₅ OH	- 52	156	0.82
cyclohexanol	cyclohexyl alcohol	cyclo-C ₆ H ₁₁ OH	25	162	0.96
1-heptanol	n-heptyl alcohol	CH ₃ (CH ₂) ₆ OH	- 34	176	0.82
1-octanol	n-octyl alcohol	CH ₃ (CH ₂) ₇ OH	- 16	194	0.83
1-nonanol	n-nonyl alcohol	CH ₃ (CH ₂) ₈ OH	- 6	214	0.83
1-decanol	n-decyl alcohol	CH ₃ (CH ₂) ₉ OH	6	233	0.83
2-propen-1-ol	allyl alcohol	H ₂ C=CH-CH ₂ OH	- 129	97	0.86
phenylmethanol	benzyl alcohol	Ph-CH ₂ OH	- 15	205	1.05
diphenylmethanol	diphenylcarbinol	Ph ₂ CHOH	69	298	
triphenylmethanol	triphenylcarbinol	Ph ₃ COH	162	380	1.20
1, 2-ethanediol	ethylene glycol	HOCH ₂ CH ₂ OH	- 13	198	1.12
1, 2-propanediol	propylene glycol	CH ₃ CH(OH)CH ₂ OH	- 59	188	1.04
1, 2, 3-propanetriol	glycerol	HOCH ₂ CH(OH)CH ₂ OH	18	290	1.26

Table 2 : Physical Properties of Some Representative Ethers

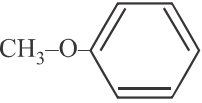
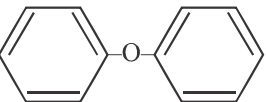
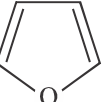
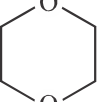
Name	Structure	MP(°C)	BP(°C)	Density (g/mL)
dimethyl ether	$\text{CH}_3\text{--O--CH}_3$	– 140	– 25	0.66
ethyl methyl ether	$\text{CH}_3\text{CH}_2\text{--O--CH}_3$		8	0.72
diethyl ether	$\text{CH}_3\text{CH}_2\text{--O--CH}_2\text{CH}_3$	– 116	35	0.71
di-n-propyl ether	$\text{CH}_3\text{CH}_2\text{CH}_2\text{--O--CH}_2\text{CH}_2\text{CH}_3$	– 122	91	0.74
diisopropyl ether	$(\text{CH}_3)_2\text{CH--O--CH}(\text{CH}_3)_2$	– 86	68	0.74
1, 2-dimethoxyethane (DME)	$\text{CH}_3\text{--O--CH}_2\text{CH}_2\text{--O--CH}_3$	– 58	83	0.86
methyl phenyl ether (anisole)		– 37	154	0.99
diphenyl ether		27	259	1.07
furan		–86	32	0.94
tetrahydrofuran (THF)		–108	65	0.89
1, 4-dioxane		11	101	1.03

Table 3 : Comparison of the Boiling Points of Ethers, Alkanes and Alcohols of Similar Molecular Weights

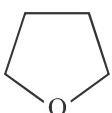
Compound	Formula	MW	BP(°C)	Dipole Moment (D)
water	H_2O	18	100	1.9
ethanol	$\text{CH}_3\text{CH}_2\text{--OH}$	46	78	1.7
dimethyl ether	$\text{CH}_3\text{--O--CH}_3$	46	– 25	1.3
propane	$\text{CH}_3\text{CH}_2\text{CH}_3$	44	– 42	0.1
n-butanol	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{--OH}$	74	118	1.7
tetrahydrofuran		72	65	1.63
diethyl ether	$\text{CH}_3\text{CH}_2\text{--O--CH}_2\text{CH}_3$	74	35	1.2
pentane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	72	36	0.1

Table 4 : Solubility of Alcohols in Water (at 25°C)

Alcohol	Solubility in Water
methyl	miscible
ethyl	miscible
n-propyl	miscible
t-butyl	miscible
isobutyl	10.0%
n-butyl	9.1%
n-pentyl	2.7%
cyclohexyl	3.6%
n-hexyl	0.6%
phenol	9.3%
hexane-1, 6-diol	miscible

Table 5 : Acid-Dissociation Constants of Representative Alcohols

Alcohol	Structure	K_a	pK_a
methanol	CH_3-OH	3.2×10^{-16}	15.5
ethanol	CH_3CH_2-OH	1.3×10^{-16}	15.9
2-chloroethanol	$Cl-CH_2CH_2-OH$	5.0×10^{-15}	14.3
isopropyl alcohol	$(CH_3)_2CH-OH$	3.2×10^{-17}	16.5
t-butyl alcohol	$(CH_3)_3C-OH$	1.0×10^{-18}	18.0
cyclohexanol	$C_6H_{11}-OH$	1.0×10^{-18}	18.0
phenol	C_6H_5-OH	1.0×10^{-10}	10.0
Comparison with other Acids			
water	H_2O	1.8×10^{-16}	15.7
acetic acid	CH_3COOH	1.6×10^{-5}	4.8
hydrochloric acid	HCl	$1.6 \times 10^{+2}$	-2.2

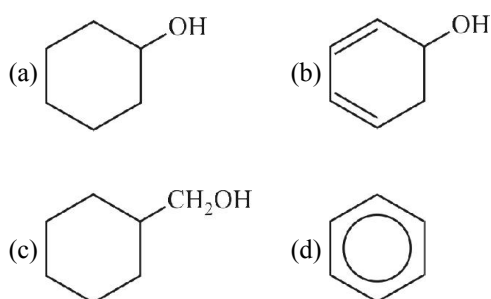
EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Properties of Alcohols

1. Glycerine has

- (a) One primary and two secondary –OH groups
- (b) One secondary and two primary –OH groups
- (c) Three primary –OH groups
- (d) Three secondary –OH groups

2. The structural formula of cyclohexanol is



3. Which of the following is not characteristic of alcohols

- (a) They are lighter than water
- (b) Their boiling points rise fairly uniformly with increasing molecular weight
- (c) Lower members are insoluble in water and organic solvents but solubility regularly increases with molecular weight
- (d) Lower members have pleasant smell and burning taste, while higher members are odourless and tasteless

Preparation of Alcohols

4. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives

- (a) o-cresol (b) p-cresol
- (c) 2, 4-dihydroxy toluene (d) Benzyl alcohol

5. Which of the following reagents convert the propene to 1-propanol

- (a) H_2O , H_2SO_4 (b) Aqueous KOH
- (c) MgSO_4 , $\text{NaBH}_4/\text{H}_2\text{O}$ (d) B_2H_6 , H_2O_2 , OH^-

6. Ethylene reacts with Baeyer's reagent to give

- (a) Ethane (b) Ethyl alcohol
- (c) Ethylene glycol (d) None of these

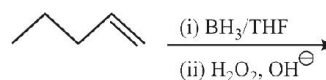
7. Propan-1-ol can be prepared from propene by alcohol

- (a) $\text{H}_2\text{O}/\text{H}_2\text{SO}_4$
- (b) $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}$ followed by NaBH_4
- (c) B_2H_6 followed by H_2O_2
- (d) $\text{CH}_3\text{CO}_2\text{H}/\text{H}_2\text{SO}_4$

8. Oxymercuration-demercuration reaction of 1-methylcyclohexene gives

- (a) trans-2-methyl cyclohexanol
- (b) cis-2-methylcyclohexanol
- (c) 1-methylcyclohexanol
- (d) mixture of cis- and trans-2-methylcyclohexanol

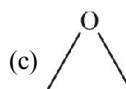
9. The product of the following reaction is



- (a) 2-pentanol (b) pentane
- (c) pentan-2-one (d) 1-pentanol

10. 2-phenylethanol may be prepared by the reaction of phenylmagnesium bromide with

- (a) HCHO (b) CH_3COCH_3



- (d) CH_3CHO

Reactions of Alcohols

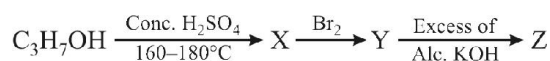
11. The alcohol that produces turbidity immediately with $\text{ZnCl}_2 + \text{conc. HCl}$ at room temperature

(a) 1-hydroxybutane
(b) 2-hydroxybutane
(c) 2-hydroxy-2-methylpropane
(d) 1-hydroxy-2-methylpropane

12. During dehydration of alcohols to alkenes by heating with conc. H_2SO_4 the initiation step is

(a) Protonation of alcohol molecule
(b) Formation of carbocation
(c) Elimination of water
(d) Formation of an ester

13. In the following series of chemical reactions, identify Z



(a) $\text{CH}_3-\underset{\text{NH}_2}{\text{CH}}-\underset{\text{NH}_2}{\text{CH}_2}$ (b) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\underset{\text{OH}}{\text{CH}_2}$
(c) $\text{CH}_3-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ (d) $\text{CH}_3\text{C}\equiv\text{CH}$

14. Lucas test is used for

(a) Alcohols (b) Amines
(c) Diethyl ether (d) Glacial acetic acid

15. Compound 'A' reacts with PCl_5 to give 'B' which on treatment with KCN followed by hydrolysis gave propanoic acid as the product. What is 'A'

(a) Ethane (b) Propane
(c) Ethyl chloride (d) Ethyl alcohol

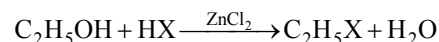
16. Lucas reagent is

(a) Conc. HCl and anhydrous ZnCl_2
(b) Conc. HNO_3 and hydrous ZnCl_2
(c) Conc. HCl and hydrous ZnCl_2
(d) Conc. HNO_3 and anhydrous ZnCl_2

17. HBr reacts fastest with

(a) 2-Methylpropan-1-ol (b) 2-Methylpropan-2-ol
(c) Propan-2-ol (d) Propan-1-ol

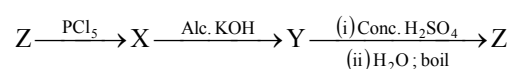
18. For the reaction,



the order of reactivity is

(a) $\text{HBr} > \text{HI} > \text{HCl}$ (b) $\text{HI} > \text{HCl} > \text{HBr}$
(c) $\text{HI} > \text{HBr} > \text{HCl}$ (d) $\text{HCl} > \text{HBr} > \text{HI}$

19. What is Z in the following sequence of reactions ?



(a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (b) $\text{CH}_3\text{CHOHCH}_3$
(c) $(\text{CH}_3\text{CH}_2)_2\text{CHOH}$ (d) $\text{CH}_3\text{CH}=\text{CH}_2$

20. Methanol is industrially prepared by

(a) Oxidation of CH_4 by steam at 900°C
(b) Reduction of HCHO using LiAlH_4
(c) Reaction HCHO with a solution of NaOH
(d) Reduction of CO using H_2 and $\text{ZnO-Cr}_2\text{O}_3$

21. Victor Meyer's test is not given by

(a) $(\text{CH}_3)_3\text{COH}$ (b) $\text{C}_2\text{H}_5\text{OH}$
(c) $(\text{CH}_3)_2\text{CHOOH}$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

22. An organic compound A reacts with sodium metal and forms B. On heating with conc. H_2SO_4 , A gives diethyl ether. A and B are respectively

(a) $\text{C}_2\text{H}_5\text{OH}$ and $\text{C}_2\text{H}_5\text{ONa}$ (b) $\text{C}_3\text{H}_7\text{OH}$ and $\text{C}_3\text{H}_7\text{ONa}$
(c) CH_3OH and CH_3ONa
(d) $\text{C}_4\text{H}_9\text{OH}$ and $\text{C}_4\text{H}_9\text{ONa}$

23. Conc. H_2SO_4 reacts with $\text{C}_2\text{H}_5\text{OH}$ at 170°C to form

(a) CH_3COCH_3 (b) CH_3COOH
(c) CH_3CHO (d) C_2H_4

24. Isopropyl alcohol heated at 300°C with copper catalyst to form

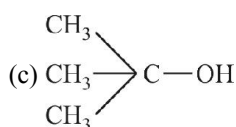
- (a) Acetone (b) Dimethyl ether
(c) Acetaldehyde (d) Ethane

25. Ethylene may be obtained by dehydration of which of the following with concentrated H_2SO_4 at $160-170^{\circ}\text{C}$

- (a) $\text{C}_2\text{H}_5\text{OH}$ (b) CH_3OH
(c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (d) $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$

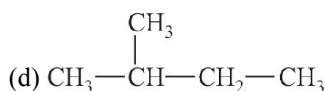
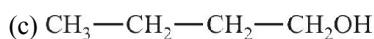
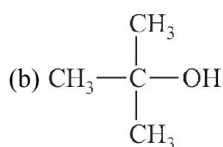
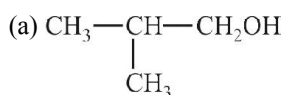
26. Which of the following vapours passed over heated copper to form acetone

- (a) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2\text{OH}$
(b) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$



- (d) $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$

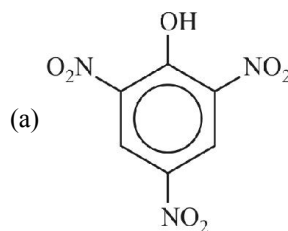
27. The compound which gives the most stable carbonium ion on dehydration is



28. Alcohols react with Grignard reagent to form

- (a) Alkanes (b) Alkenes
(c) Alkynes (d) All of these

29. Which of the following will not react with NaOH

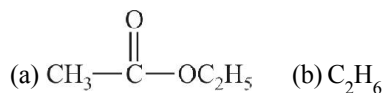


- (b) $\text{C}_2\text{H}_5\text{OH}$
(c) CH_3CONH_2 (d) $\text{CH}(\text{CN})_3$

30. Amongst the following, HBr reacts fastest with

- (a) Propane-1-ol (b) Propane-2-ol
(c) 2-methyl propane-1-ol (d) 2-methyl propane-2-ol

31. Ethyl alcohol is heated with conc. H_2SO_4 . The product formed is



- (c) C_2H_4 (d) C_2H_2

32. Which will dehydrate most easily

- (a) 3-methyl-2-butanol (b) Ethyl alcohol
(c) 2-methyl propane-2-ol (d) 2-methyl butan-2-ol

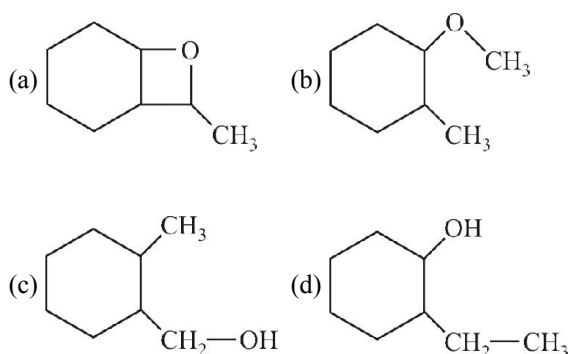
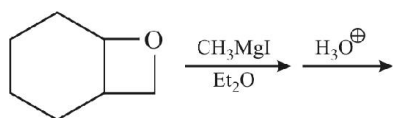
33. The best method to prepare cyclohexene from cyclohexanol is by using

- (a) Conc. $\text{HCl} + \text{ZnCl}_2$ (b) Conc. H_3PO_4
(c) HBr (d) Conc. HCl

34. An organic compound 'A' reacts with methyl magnesium iodide, to form an addition product which on hydrolysis forms the compound 'B'. Compound 'B' gives blue colour salt in Victor Meyer's test. The compounds 'A' and 'B' are respectively

- (a) acetaldehyde and isopropyl alcohol
(b) acetone and isopropyl alcohol
(c) acetaldehyde and tertiary butyl alcohol
(d) acetaldehyde and ethyl alcohol

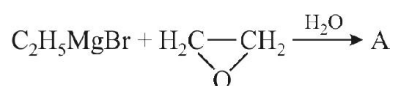
35. What is the major product of the following reaction ?



Ethers

Preparation and Properties

36. In the following reaction 'A' is



- (a) $\text{C}_2\text{H}_5\text{CH}_2\text{CHO}$ (b) $\text{C}_2\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$
 (c) $\text{C}_2\text{H}_5\text{CH}_2\text{OH}$ (d) $\text{C}_2\text{H}_5\text{CHO}$

37. Ether is obtained from ethyl alcohol

- (a) in presence of H_2SO_4 at 413 K
 (b) in presence of H_2SO_4 at 474 K
 (c) in presence of H_2SO_4 at 383 K
 (d) in presence of H_2SO_4 at 273 K

38. Diethyl ether absorbs oxygen to form

- (a) Red coloured sweet smelling compound
 (b) Acetic acid
 (c) Ether suboxide (d) Ether peroxide

39. An ether is more volatile than an alcohol having the same molecular formula. This is due to

- (a) Dipolar character of ethers
 (b) Alcohols have resonating structures
 (c) Inter-molecular hydrogen bonding in ethers
 (d) Inter-molecular hydrogen bonding in alcohols

40. The compound which does not react with sodium is

- (a) $\text{C}_2\text{H}_5\text{OH}$ (b) $\text{CH}_3-\text{O}-\text{CH}_3$
 (c) CH_3COOH (d) $\text{CH}_3-\text{CHOH}-\text{CH}_3$

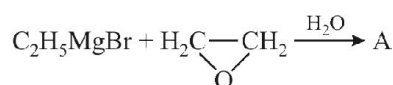
41. Product C in the following reaction, will be



- (a) ethane (b) ethyl methyl ether
 (c) ethyl iodide (d) propane

Reactions of Ether

42. In the following reaction 'A' is



- (a) $\text{C}_2\text{H}_5\text{CH}_2\text{CHO}$ (b) $\text{C}_2\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$
 (c) $\text{C}_2\text{H}_5\text{CH}_2\text{OH}$ (d) $\text{C}_2\text{H}_5\text{CHO}$

43. Which of the following product is formed, when ether is exposed to air

- (a) Oxide (b) Alkanes
 (c) Alkenes (d) Peroxide of diethyl ether

44. Diethyl ether is heated with one mole of HI, which is formed

- (a) Ethyl alcohol and ethyl iodide
 (b) Ethyl iodide only
 (c) Ethyl alcohol only (d) Ethyl iodide and ethane

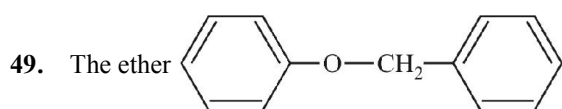
45. On boiling with concentrated hydrobromic acid, phenyl ethyl ether will yield

- (a) Phenol and ethyl bromide
 (b) Bromobenzene and ethanol
 (c) Phenol and ethane
 (d) Bromobenzene and ethane

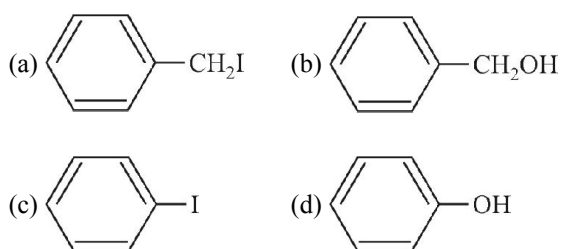
46. The reaction of $\text{CH}_2=\text{CH}_2$ with RMgX leads to the formation of

- (a) RCHOHR (b) RCHOHCH_3
 (c) $\text{R}_2\text{CHCH}_2\text{OH}$ (d) $\text{RCH}_2\text{CH}_2\text{OH}$

47. Diethyl ether absorbs oxygen to form
- Red coloured sweet smelling compound
 - Acetic acid
 - Ether suboxide
 - Ether peroxide
48. On boiling with concentrated hydrobromic acid, phenyl ethyl ether will yield
- Phenol and ethyl bromide
 - Phenol and ethane
 - Bromobenzene and ethanol
 - Bromobenzene and ethane

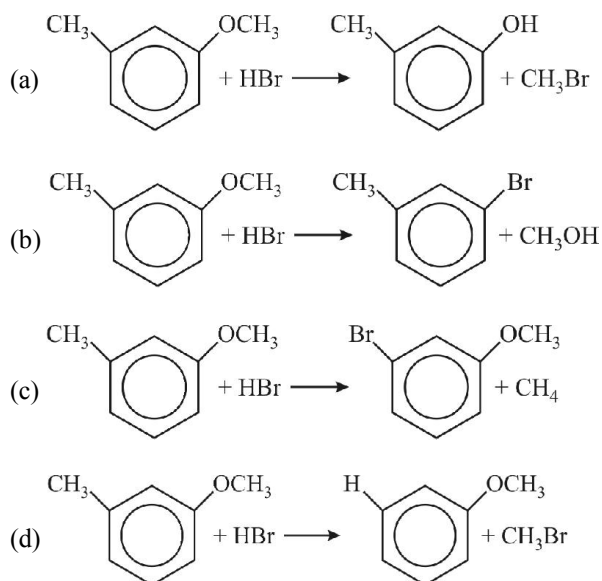


when treated with HI produces



50. The ether that undergoes electrophilic substitution reactions is
- $\text{CH}_3\text{OC}_2\text{H}_5$
 - $\text{C}_6\text{H}_5\text{OCH}_3$
 - CH_3OCH_3
 - $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
51. The products formed in the following reaction
- $$\text{C}_6\text{H}_5-\text{O}-\text{CH}_3 + \text{HI} \xrightarrow{\text{heat}} \text{are}$$
- $\text{C}_6\text{H}_5-\text{I}$ and CH_3-OH
 - $\text{C}_6\text{H}_5-\text{OH}$ and CH_3-I
 - $\text{C}_6\text{H}_5-\text{CH}_3$ and HOI
 - C_6H_6 and CH_3OI

52. Which of the following reaction is correctly represented



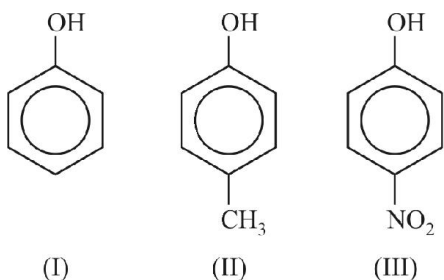
53. An organic compound of molecular formula $\text{C}_4\text{H}_{10}\text{O}$ does not react with sodium. With excess of HI, it gives only one type of alkyl halide. The compound is
- 1-butanol
 - ethoxyethane
 - 1-methoxypropane
 - 2-methoxypropane

Phenols

Preparation and Properties

54. Ortho-nitrophenol is steam volatile whereas para-nitrophenol is not. This is due to
- Intramolecular hydrogen bonding present in ortho-nitrophenol
 - Intermolecular hydrogen bonding
 - Intramolecular hydrogen bonding present in para-nitrophenol
 - None of these
55. Phenol is less acidic than
- Acetic acid
 - p-nitrophenol
 - Both (a) and (b)
 - None of these
56. Phenol is less acidic than
- Ethanol
 - Methanol
 - o-nitrophenol
 - p-methylphenol

57. The increasing order of acidity among phenol, p-methylphenol, m-nitrophenol and p-nitrophenol is
- (a) m-nitrophenol, p-nitrophenol, phenol, p-methylphenol
(b) p-methylphenol, m-nitrophenol, phenol, p-nitrophenol
(c) p-methylphenol, phenol, m-nitrophenol, p-nitrophenol
(d) Phenol, p-methylphenol, p-nitrophenol, m-nitrophenol
58. Increasing order of acid strength among p-methoxyphenol, p-methylphenol and p-nitrophenol is
- (a) p-Nitrophenol, p-Methoxyphenol, p-Methylphenol
(b) p-Methylphenol, p-Methoxyphenol, p-Nitrophenol
(c) p-Nitrophenol, p-Methylphenol, p-Methoxyphenol
(d) p-Methoxyphenol, p-Methylphenol, p-Nitrophenol
59. The correct acidic order of the following is

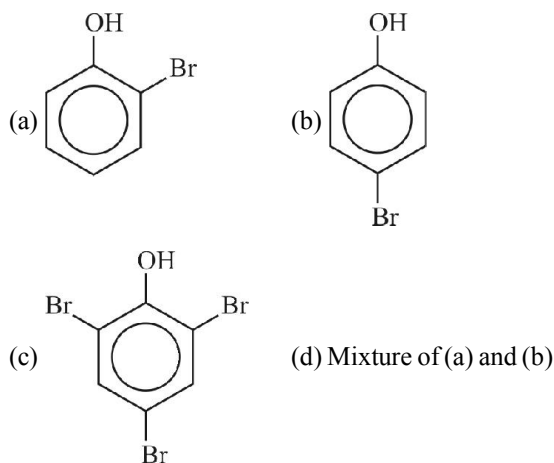


- (a) I > II > III (b) III > I > II
(c) II > III > I (d) I > III > II
60. Picric acid is
- (a) Trinitrotoluene (b) Trinitroaniline
(c) A volatile liquid (d) 2, 4, 6-Trinitrophenol

Reactions of Phenols

61. Kolbe-Schmidt reaction is used for
- (a) Salicylic acid (b) Salicylaldehyde
(c) Phenol (d) Hydrocarbon
62. Benzenediazonium chloride on reaction with phenol in weakly basic medium gives
- (a) Diphenyl ether (b) p-hydroxyazobenzene
(c) Chlorobenzene (d) Benzene

63. With excess bromine, phenol reacts to form



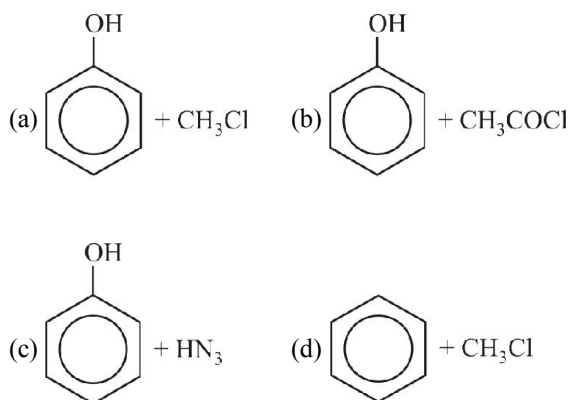
64. Which of the following is a Reimer-Tiemann reaction

- (a) $\text{C}_6\text{H}_5\text{OH} + \text{CCl}_4 \xrightarrow{\text{NaOH}}$ Salicylic acid
(b) $\text{C}_6\text{H}_5\text{OH} + \text{Zn} \longrightarrow \text{C}_6\text{H}_6$
(c) $\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \longrightarrow \text{C}_6\text{H}_5\text{ONa}$
(d) None of these

65. In presence of NaOH, phenol react with CHCl_3 to form o-hydroxy benzaldehyde. This reaction is called

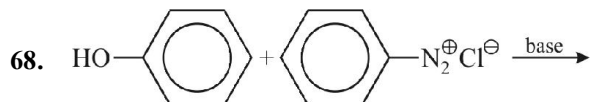
- (a) Reimer-Tiemann's reaction
(b) Sandmeyer's reaction
(c) Hoffmann's degradation reaction
(d) Gattermann's aldehyde synthesis

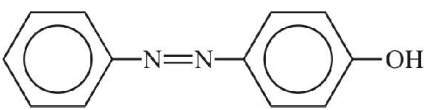
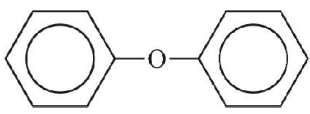
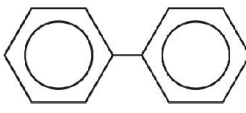
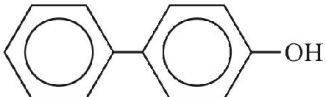
66. In Friedal-Crafts acylation, besides AlCl_3 , the other reactants are

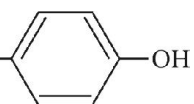


67. Which of the following reagents will produce salicylaldehyde on reaction with phenol

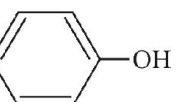
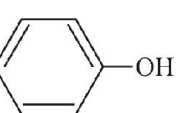
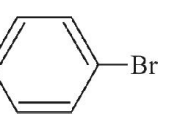
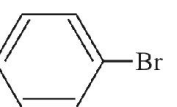
- (a) $\text{CHCl}_3/\text{NaOH}$ (b) CCl_4/NaOH
(c) $\text{CH}_2\text{Cl}_2/\text{NaOH}$ (d) $\text{CH}_3\text{Cl}/\text{NaOH}$



- (a) 
(b) 
(c) 
(d) 

69. The reaction of $\text{CH}_3\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ 

with HBr gives

- (a) $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{OH}$ 
(b) $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{OH}$ 
(c) $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_3(\text{Br})-\text{OH}$ 
(d) $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_3(\text{Br})-\text{OH}$ 

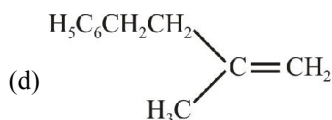
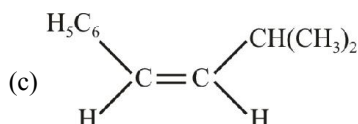
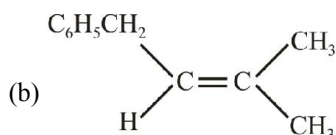
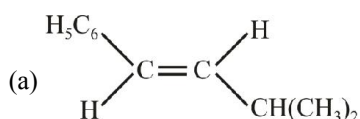
70. The compound 'A' when treated with methyl alcohol and few drops of H_2SO_4 gave smell of winter green. The compound 'A' is

- (a) oxalic acid (b) tartaric acid
(c) salicylic acid (d) succinic acid

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

1. During dehydration of alcohols to alkenes by heating with concentrated H_2SO_4 the initiation step is (2002)
- (a) protonation of alcohol molecule
(b) formation of carbocation
(c) elimination of water
(d) formation of an ester
2. An ether is more volatile than an alcohol having the same molecular formula. This is due to (2003)
- (a) dipolar character of ethers
(b) alcohols having resonance structures
(c) intermolecular hydrogen bonding in ethers
(d) intermolecular hydrogen bonding in alcohols
3. Among the following compounds which can be dehydrated very easily ? (2004)
- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
(c) $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)(\text{OH})\text{CH}_2\text{CH}_3$ (d) $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{OH}$
4. The best reagent to convert pent-3-en-2-ol into pent-3-en-2-one is (2005)
- (a) MnO_2
(b) chromic anhydride in glacial acetic acid
(c) acidic dichromate
(d) acidic permanganate
5. Acid catalyzed hydration of alkenes except ethene leads to the formation of (2005)
- (a) mixture of secondary and tertiary alcohols
(b) mixture of primary and secondary alcohols
(c) secondary or tertiary alcohol
(d) primary alcohol
6. The structure of the compound that gives a tribromo derivative on treatment with bromine water is (2006)
- (a)
- (b)
- (c)
- (d)
- 7.
- The electrophile involved in the above reaction is (2006)
- (a) dichloromethyl cation (CHCl_2^+)
(b) dichlorocarbene ($:\text{CCl}_2$)
(c) trichloromethyl anion (CCl_3^-)
(d) formyl cation (CHO^+)
8. Phenyl magnesium bromide reacts with methanol to give (2006)
- (a) a mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$
(b) a mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
(c) a mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$
(d) a mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$
9. In the following sequence of reactions, (2007)
- $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow{\text{P} + \text{I}_2} \text{A} \xrightarrow[\text{ether}]{\text{Mg}} \text{B} \xrightarrow{\text{HCHO}} \text{C} \xrightarrow{\text{H}_2\text{O}} \text{D}$
- the compound 'D' is
- (a) butanal
(b) n-butyl alcohol
(c) n-propyl alcohol
(d) propanal

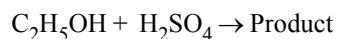
10. Phenol, when it first reacts with concentrated sulphuric acid and then with concentrated nitric acid, gives (2008)
(a) 2, 4, 6-trinitrobenzene (b) o-nitrophenol
(c) p-nitrophenol (d) nitrobenzene
11. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was (2008)
(a) CH_3OH (b) HCHO
(c) CH_3COCH_3 (d) CH_3COOH
12. The main product of the following reaction is
 $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{CH}_3)_2 \xrightarrow{\text{Conc. H}_2\text{SO}_4} ?$ (2009)



13. From amongst the following alcohols the one that would react fastest with conc HCl and anhydrous ZnCl_2 , is (2010)

- (a) 2-butanol (b) 2-methylpropan-2-ol
(c) 2-methylpropanol (d) 1-butanol

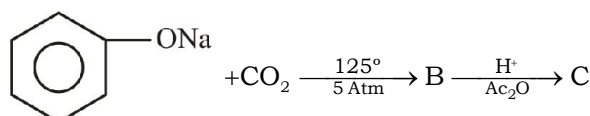
14. Consider the following reaction



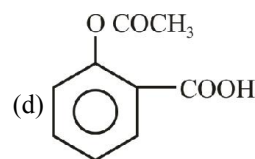
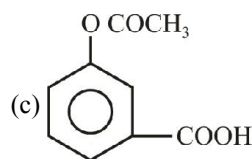
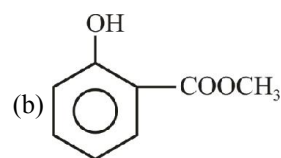
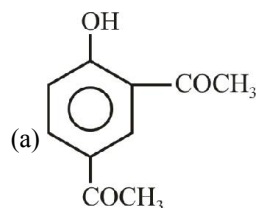
Among the following, which one cannot be formed as a product under any conditions? (2010)

- (a) Ethyl hydrogen sulphate
(b) Ethylene
(c) Acetylene
(d) Diethyl ether

15. Which of the following reagents may be used to distinguish between phenol and benzoic acid? (2011)
(a) Aqueous NaOH (b) Tollen's reagent
(c) Molisch reagent (d) Neutral FeCl_3
16. Phenol is heated with a solution of mixture of KBr and KBrO_3 . The major product obtained in the above reaction is (2011)
(a) 2-bromophenol (b) 3-bromophenol
(c) 4-bromophenol (d) 2, 4, 6-tribromophenol
17. Sodium phenoxide when heated with CO_2 under pressure at 125°C yields a product which on acetylation produces C. (2014)

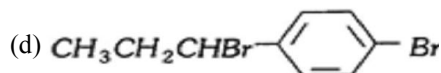
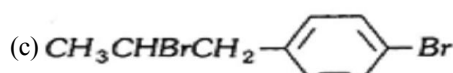
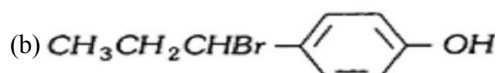
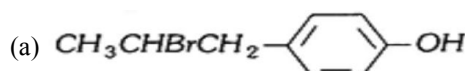


The major product C would be :

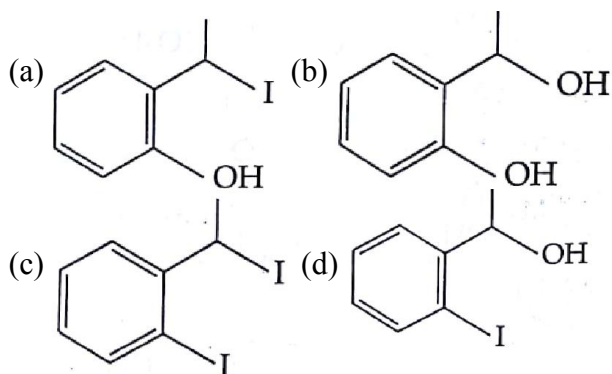
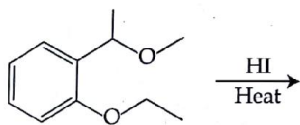


18. The most suitable reagent for the conversion of $\text{R}-\text{CH}_2-\text{OH} \rightarrow \text{R}-\text{CHO}$ is : (2014)
(a) $\text{K}_2\text{Cr}_2\text{O}_7$ (b) CrO_3
(c) PCC (Pyridinium Chlorochromate)
(d) KMnO_4

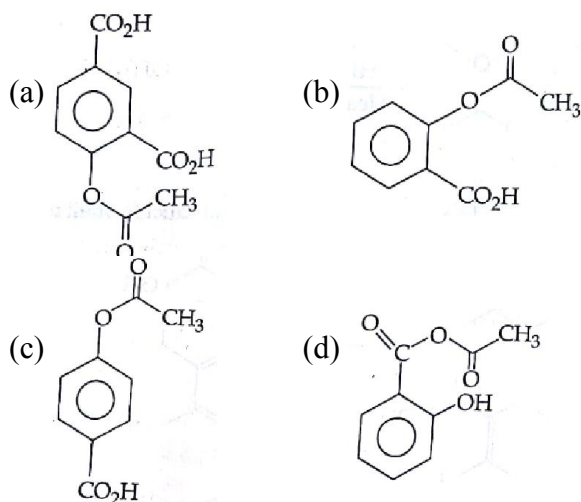
19. The reaction of $\text{CH}_3\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ with HBr gives (2015)



20. The major product formed in the following reaction is: (2017)



21. Phenol on treatment with CO_2 in the presence of NaOH followed by acidification produces compound X as the major product. X on treatment with $(\text{CH}_3\text{CO})_2\text{O}$ in the presence of catalytic amount of H_2SO_4 produces. (2018)



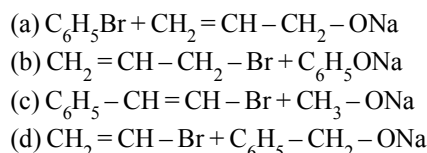
JEE MAINS ONLINE QUESTION

1. In the hydroboration-oxidation reaction of propene with diborane, H_2O_2 and NaOH , the organic compound formed is: **Online 2014 SET (1)**



2. Allyl phenyl ether can be prepared by heating:

Online 2014 SET (1)



3. Which one of the following statements is not correct?

Online 2014 SET (2)

- (a) Alcohols are weaker acids than water
(b) Carbon-oxygen bond length in methanol, CH_3OH is shorter than that of C-O bond length in phenol.
(c) Acid strength of alcohols decreases in the following order $\text{RCH}_2\text{OH} > \text{R}_2\text{CHOH} > \text{R}_3\text{COH}$

- (d) The bond angle in methanol is 108.9°

4. In the Victor-Meyer's test, the colour given by 1° , 2° and 3° alcohols are respectively: **Online 2014 SET (3)**

- (a) Red, blue, violet (b) Colourless, red, blue
(c) Red, colourless, blue
(d) Red, blue, colourless

5. Phthalic acid reacts with resorcinol in the presence of concentrated H_2SO_4 to give; **Online 2014 SET (3)**

- (a) Phenolphthalein (b) Fluorescein
(c) Alizarin (d) Coumarin

6. Williamson synthesis of ether is an example of :

Online 2014 SET (4)

- (a) Electrophilic addition
(b) Nucleophilic addition
(c) Electrophilic substitution
(d) Nucleophilic substitution

7. Phenol, when first reacts with concentrated sulphuric acid and then concentrated nitric acid, gives

Online 2016 SET (1)

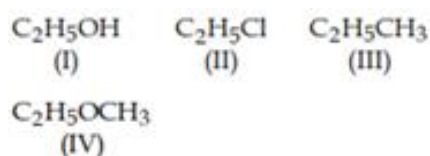
- (a) o-nitrophenol (b) p-nitrophenol
(c) Nitrobenzene (d) 2, 4, 6-trinitrophenol

8. The correct statement about the synthesis of erythritol ($C(CH_2OH)_4$) used in the preparation of PETN is :

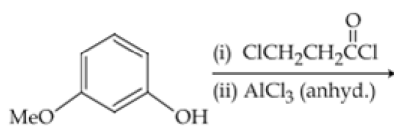
Online 2016 SET (2)

- (a) The synthesis requires four aldol condensations between methanol and ethanol.
(b) The synthesis requires two aldol condensations and two Cannizzaro reactions.
(c) The synthesis requires three aldol condensations and one Cannizzaro reaction.
(d) Alpha hydrogens of ethanol and methanol are involved in this reaction.

9. The increasing order of the boiling points for the following compounds is :
Online 2017 SET (2)



- (a) (III) < (IV) < (II) < (I) (b) (IV) < (III) < (I) < (II)
(c) (II) < (III) < (IV) < (I) (d) (III) < (II) < (I) < (IV)
10. The major product of the following reaction is :
Online 2018 SET (1)



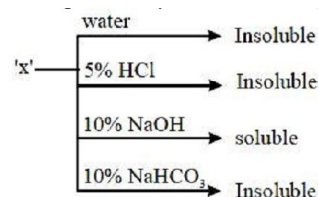
- (a)
(b)
(c)
(d)

11. Which of the following compounds will most readily be dehydrated to give alkene under acidic condition ?

Online 2018 SET (3)

- (a) 1-Pentanol
(b) 4-Hydroxypentan-2-one
(c) 3-Hydroxypentan-2-one
(d) 2-Hydroxycyclopentanone

12. An organic compound 'X' showing the following solubility profile is :
(Online 2019 SET (1))



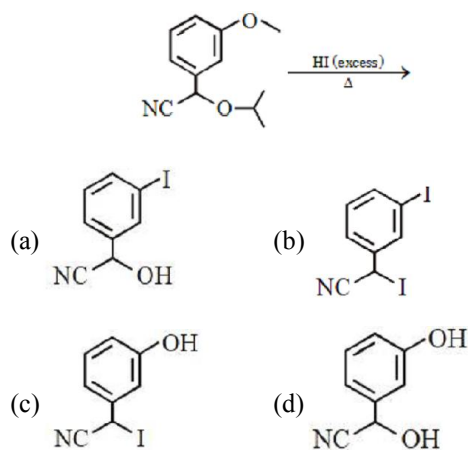
- (a) Oleic acid (b) o-Toluidine
(c) Benzamide (d) m-Cresol

13. The organic compound that gives following qualitative analysis is :
(Online 2019 SET (2))

Test	Inference
(A) Dil. HCl	Insoluble
(B) NaOH solution	Soluble
(C) Br ₂ /water	Decolourization

- (a)
(b)
(c)
(d)

14. The major product of the following reaction is :
(Online 2019 SET (3))



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

- All questions marked “S” are single choice questions
- All questions marked “M” are multiple choice questions
- All questions marked “C” are comprehension based questions
- All questions marked “A” are assertion–reason type questions
 - If both assertion and reason are correct and reason is the correct explanation of assertion.
 - If both assertion and reason are true but reason is not the correct explanation of assertion.
 - If assertion is true but reason is false.
 - If reason is true but assertion is false.
- All questions marked “X” are matrix–match type questions
- All questions marked “I” are integer type questions

Alcohols Properties

- (S) The general molecular formula, which represents the homologous series of alkanol is:
 - $C_nH_{2n}O$
 - $C_nH_{2n+1}O$
 - $C_nH_{2n+2}O$
 - $C_nH_{2n}O_2$
- (S) Total number of isomeric alcohols with formula $C_4H_{10}O$ is
 - 2
 - 1
 - 3
 - 4
- (S) Which of the following is soluble in water?
 - CH_3OH
 - $CHCl_3$
 - CCl_4
 - CS_2
- (S) Reason for excessive solubility of ethyl alcohol in water is due to:
 - covalent bonding
 - ionic bonding
 - hydrogen bonding with water
 - none of the above
- (S) Alcoholic fermentation is brought by:
 - CO_2
 - $NaHCO_3$
 - yeast
 - bacteria
- (S) Grain alcohol is common name of:
 - amyl alcohol
 - ethyl alcohol
 - methanol
 - none of these
- (S) In cold countries, ethylene glycol is added in the water used in the radiators of cars during winters. This results in:
 - lowering in freezing point
 - reducing the viscosity
 - reducing the specific heat
 - making water a better conductor of electricity
- (A) **Assertion:** Solubility of n-alcohols in water decreases with increase in molecular weight.
Reason: The relative proportion of the hydrocarbon part in alcohol increases with increasing molecular weight which permits enhanced hydrogen bond with water.
 - A
 - B
 - C
 - D
- (A) **Assertion:** The acidity of alcohols follows the order $1^\circ > 2^\circ > 3^\circ$.
Reason: The +I effect of alkyl groups ($3^\circ > 2^\circ > 1^\circ$) favours the dissociation of –O–H group.
 - A
 - B
 - C
 - D

10. (A) **Assertion:** Alcohols can be dried over anhydrous CaCl_2 .

Reason: Anhydrous CaCl_2 absorbs moisture of aqueous ethyl alcohol

- (a) A (b) B
(c) C (d) D

Preparation of Alcohols

11. (S) When CH_3MgI is made to react with acetone and the addition product is hydrolysed, we get:

- (a) primary alcohol (b) secondary alcohol
(c) tertiary alcohol (d) an aldehyde

12. (S) To prepare propan-2-ol from methyl magnesium iodide, the chemical reagent required is:

- (a) CH_3CHO (b) HCHO
(c) CH_3COCH_3 (d) CO_2

13. (S) Which of the following reagents can convert acetic acid into ethanol?

- (a) $\text{Sn} + \text{HCl}$ (b) $\text{H}_2 + \text{Pt}$
(c) $\text{LiAlH}_4 + \text{ether}$ (d) $\text{Na} + \text{alcohol}$

14. (S) Alkenes are converted into alcohols by:

- (a) hydrolysis with dil. H_2SO_4
(b) hydration of alkene with alkaline KMnO_4
(c) hydrolysis with water vapours and conc. H_2SO_4
(d) hydration of alkene with aq. KOH

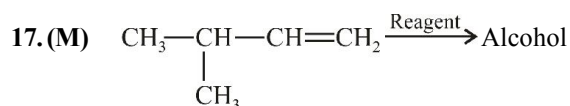
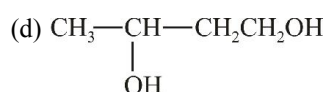
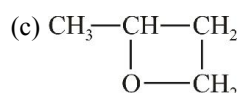
15. (S) Which of the following will produce a primary alcohol by reacting with CH_3MgI ?

- (a) Ethylene oxide (b) Ethyl acetate
(c) Methyl cyanide (d) Acetone

16. (S) The major product formed in the following reaction is:

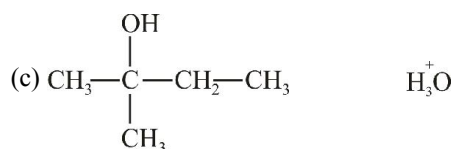
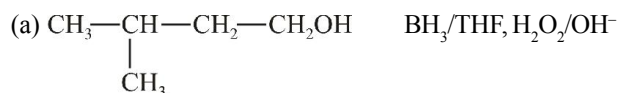


- (a) $\text{CH}_3\text{CH}=\text{CHCH}_2\text{OH}$ (b) $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{OH}$



which is/are true about alcohol and R?

Alcohol **Reagent(R)**

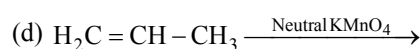
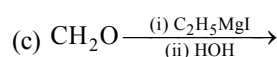


(d) None of the above

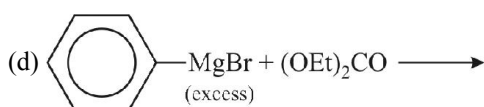
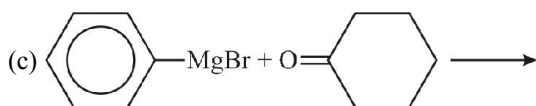
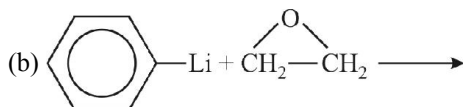
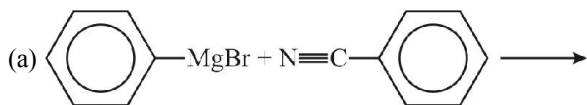
18. (M) Which of the following alcohol(s) can be prepared by the action of Grignard reagents with aldehydes?

- (a) Primary alcohols
(b) Secondary alcohols
(c) Tertiary alcohols
(d) Both primary and tertiary alcohols

19. (M) Which of the following reactions will yield propan-2-ol?



20. (M) Which of the following reactions yield alcohol as the major product?

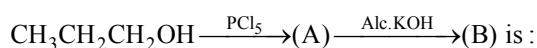


Reaction of Alcohols

21. (S) The product of reaction $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[\text{Cu}]{300^\circ\text{C}} ?$ is:

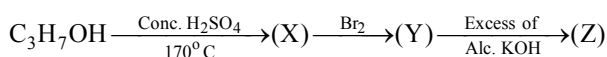
- (a) C_2H_6 (b) CH_3COCH_3
(c) CH_3COOH (d) CH_3CHO

22. (S) The compound (B) formed in the following sequence of reactions,



- (a) propene (b) propyne
(c) propane (d) propanal

23. (C) Identify (Z) in the series,

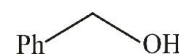
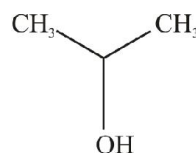
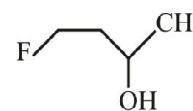
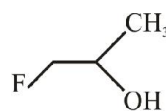


- (a) $\text{CH}_3-\text{CH}(\text{NH}_2)-\text{CH}_2(\text{NH}_2)$ (b) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_2(\text{OH})$
(c) $\text{CH}_3-\text{C}\equiv\text{CH}$ (d) $\text{CH}_3-\text{C}(\text{OH})=\text{CH}_2$

24. (S) The compound formed by reaction of ethylene glycol and periodic acid (HIO_4) is:

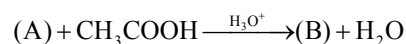
- (a) carbon dioxide (b) formic acid
(c) formaldehyde (d) glyoxal

25. (S) The order of reactivity of the following alcohols towards conc. HCl is:



- (a) $\text{I} > \text{II} > \text{III} > \text{IV}$ (b) $\text{I} > \text{III} > \text{II} > \text{IV}$
(c) $\text{IV} > \text{III} > \text{II} > \text{I}$ (d) $\text{IV} > \text{III} > \text{I} > \text{II}$

26. (S) $\text{CH}_3\text{COOH} \xrightarrow{\text{LiAlH}_4} (\text{A}),$



In the above reaction 'A' and 'B' respectively are:

- (a) $\text{CH}_3\text{COOC}_2\text{H}_5, \text{C}_2\text{H}_5\text{OH}$ (b) $\text{CH}_3\text{CHO}, \text{C}_2\text{H}_5\text{OH}$
(c) $\text{C}_2\text{H}_5\text{OH}, \text{CH}_3\text{CHO}$ (d) $\text{C}_2\text{H}_5\text{OH}, \text{CH}_3\text{COOC}_2\text{H}_5$

27. (S) In the Victor Meyer's test, the colours given by $1^\circ, 2^\circ, 3^\circ$ alcohols are respectively:

- (a) red, colourless, blue (b) red, blue, colourless
(c) colourless, red, blue (d) red, blue, violet

28. (S) A compound 'X' undergoes reduction with LiAlH_4 to yield 'Y'. When vapours of 'Y' are passed over freshly reduced copper at 300°C , 'X' is formed. What is 'Y'?

- (a) CH_3COCH_3 (b) CH_3CHO
(c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) $\text{CH}_3-\text{O}-\text{CH}_3$

29. (A) **Assertion:** HIO_4 decomposes, 1,2-glycols but not 1,3- or higher glycols.

Reason: Only 1,2-glycols from cyclic esters which subsequently undergo cleavage to form carbonyl compounds.

- (a) A (b) B
(c) C (d) D

30. (A) **Assertion:** 3° Alcohols show turbidity within 5 minutes, when treated with Lucas reagent.

Reason: Conc. HCl and anhydrous ZnCl_2 in 1:1 mixture is called Lucas reagent

- (a) A (b) B
(c) C (d) D

31. (A) **Assertion:** Grignard reagent reacts with aldehydes and ketones to form alcohol.

Reason: Alcohols have acidic hydrogen.

- (a) A (b) B
(c) C (d) D

32. (A) **Assertion :** Resorcinol turns FeCl_3 solution purple.

Reason : Resorcinol have phenolic group.

- (a) A (b) B
(c) C (d) D

33. (A) **Assertion :** A rate of hydrolysis of methyl chloride to methanol is higher in DMF than in water.

Reason : Hydrolysis of methyl chloride follows second order kinetics.

- (a) A (b) B
(c) C (d) D

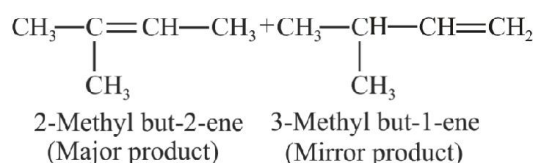
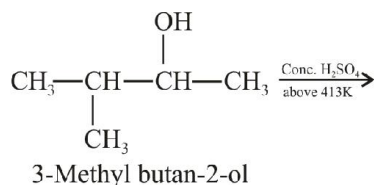
Comprehension

Secondary and tertiary alcohols always gives E_1 reaction in dehydration. Primary, alcohols whose β -carbon is 3° or 4° also give E_1 reaction. However, the primary alcohols whose β -carbon is 1° or 2° give E_2 reaction. Dehydrating agents like conc. H_2SO_4 , Al_2O_3 , anhydrous ZnCl_2 are used.

The reactivity of alcohols for elimination reaction lies in following sequence:

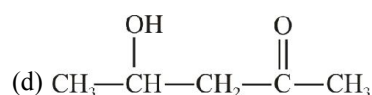
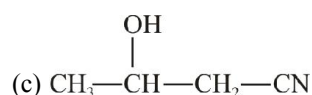
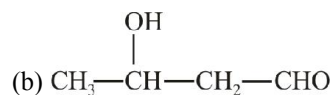
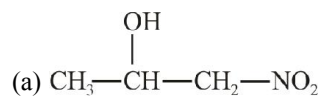
Tertiary alcohol > secondary alcohol > primary alcohol

Electron attracting groups present in alcohols increase the reactivity for dehydration. Greater is the $-I$ effect of the group present in alcohol, more will be its reactivity. Both E_1 and E_2 mechanism give the product according to Saytzeff's rule, i.e., major product is the most substituted alkene

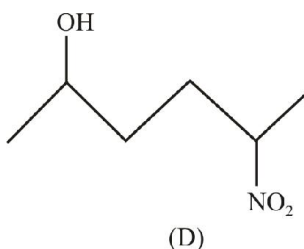
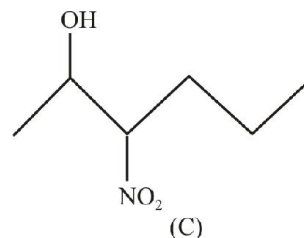
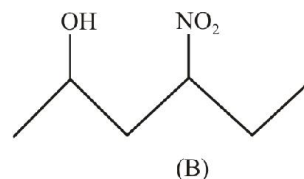
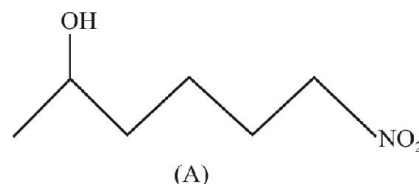


Answer the following questions:

34. (C) Which among the following alcohols will be the most reactive for dehydration?



35. (C) Arrange the reactivity of given four alcohols in decreasing order of dehydration.



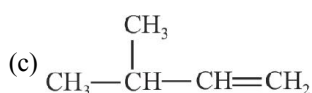
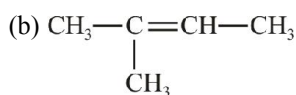
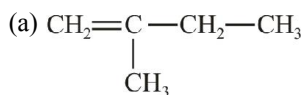
(a) $\text{A} > \text{B} > \text{C} > \text{D}$

(b) $\text{D} > \text{C} > \text{B} > \text{A}$

(c) $\text{C} > \text{B} > \text{D} > \text{A}$

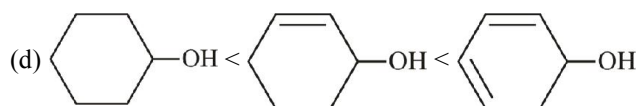
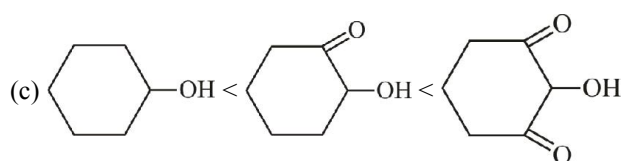
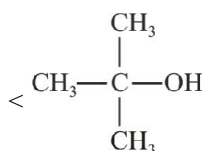
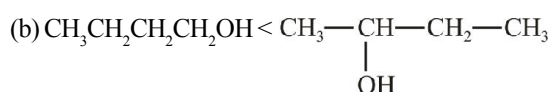
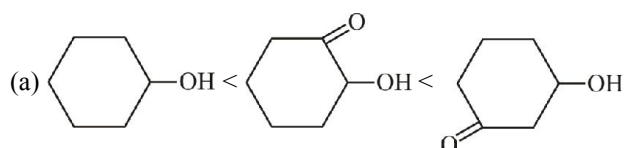
(d) $\text{B} > \text{C} > \text{A} > \text{D}$

36. (C) In the reaction, $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_2\text{OH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$ the product(s) obtained will be:

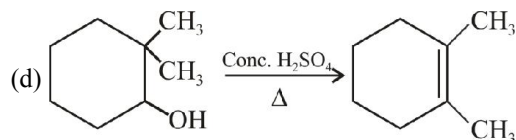
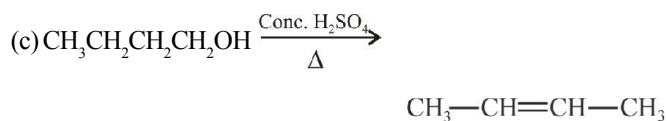
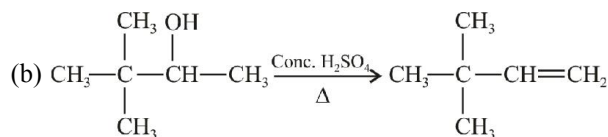
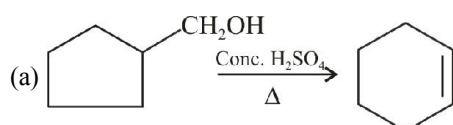


(d) this alcohol cannot be dehydrated

37. (C) Which among the following sequences of reactivity for dehydration is incorrect?



38. (C) Which of the following dehydration products is incorrect?



Ethers

39. (S) The IUPAC name of $\text{CH}_3\text{OC}_2\text{H}_5$ is:

- (a) ethoxy methane (b) methoxy ethane
(c) ethyl methyl ether (d) methyl ethyl ether

40. (S) Heating together sodium ethoxide and ethyl chloride will give:

- (a) ether (b) ethyl alcohol
(c) acetaldehyde (d) acetic acid

41. (S) According to Lewis concept of acids and bases, ethers are:

- (a) neutral (b) acidic
(c) basic (d) amphoteric

42. (S) $\text{CH}_2=\text{CH}_2 + \text{LiAlH}_4 \longrightarrow (\text{A})$, (A) is:

- (a) CH_3CHO (b) $\text{CH}_3\text{CH}_2\text{OH}$
(c) $\text{H}_2\text{C}=\text{CH}_2$ (d) $\text{CH}_2\text{OH}-\text{CH}_2\text{OH}$

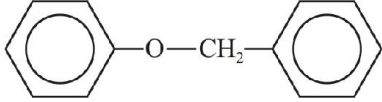
43. (S) Methyl phenyl ether can be obtained by reacting:

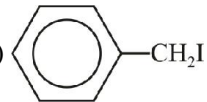
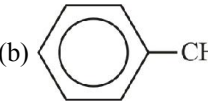
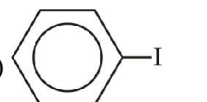
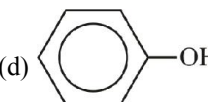
- (a) phenolate ions and methyl iodide
(b) methoxide ion and bromobenzene
(c) methanol and phenol
(d) bromobenzene and methyl bromide

44. (S) From Williamson's synthesis, the preparation of which of the following is possible?

- (a) Only symmetrical ethers
(b) Only asymmetrical ethers
(c) Both (a) and (b)
(d) None of the above

45. (S) Williamson's synthesis involves:
- (a) S_{N1} mechanism (b) nucleophilic addition
(c) S_{N2} mechanism (d) S_E mechanism
46. (M) Diethyl ether can be distinguished from n-butanol by:
- (a) aqueous $FeCl_3$
(b) reaction with Na metal
(c) Tollens reagent
(d) reaction with CrO_3 in dil H_2SO_4

47. (M) The ether  when treated with HI produces:

- (a)  (b) 
(c)  (d) 

48. (M) $C_6H_5-O-CH_3$ is named as:
- (a) phenoxy methane (b) methoxy benzene
(c) anisole (d) methyl phenyl ether
49. (M) Indicate the correct statement for diethyl ether that it is:
- (a) used as an anaesthetic (b) used as refrigerant
(c) decomposed by HI (d) used as an inert solvent
50. (M) What is the function of diethyl ether in the preparation of Grignard reagent?
- (a) To act as a catalyst
(b) To act as a solvent
(c) To provide lone pair of electrons for coordination
(d) To act as an acid

Comprehension

By a proper choice of reagents, both symmetrical and unsymmetrical ethers can be prepared by Williamson synthesis which involves the reaction between an alkyl halide and an alkoxide ion. The reverse process involving the cleavage of ethers to give back the original alkyl halide and the alcohol can be carried out by heating the ether with HI at 373 K.

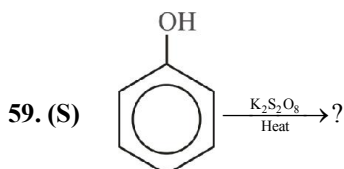
51. (C) Which of the following reagents when heated will give a good yield of an ether
- (a) Isopropyl bromide and sodium isopropoxide
(b) Isopropyl bromide and sodium ethoxide
(c) Bromobenzene and sodium phenoxide
(d) Sodium tert-butoxide and ethyl bromide
52. (C) Which of the following ethers can be prepared by Williamson synthesis ?
- (a) Benzyl methyl ether (b) Methyl vinyl ether
(c) Divinyl ether (d) Diphenyl ether
53. (C) Allyl phenyl ether can be prepared by heating
- (a) $C_6H_5Br + CH_2 = CH-CH_2ONa$
(b) $CH_2 = CHCH_2Br + C_6H_5ONa$
(c) $C_6H_5CH = CHBr + CH_3ONa$
(d) $CH_2 = CHBr + C_6H_5CH_2ONa$
54. (C) Benzyl ethyl ether reacts with HI to form
- (a) p-Iodotoluene and ethyl alcohol
(b) Benzyl alcohol and ethyl iodide
(c) Benzyl iodide and ethyl alcohol
(d) Iodobenzene and ethyl alcohol

Phenols

55. (S) Cumene $\xrightarrow[(ii) H_2O, H^+]{(i) O_2}$ (X) and (Y) :
- (X) and (Y) respectively are:
- (a) toluene, propene (b) toluene, propylchloride
(c) phenol, acetone (d) phenol, acetaldehyde
56. (S) Phenol $\xrightarrow{\text{Zinc distillation}}$ (A) $\xrightarrow[\text{at } 60^\circ C]{\text{Conc. } HNO_3, \text{ Conc. } H_2SO_4}$ (B) $\xrightarrow[NaOH]{Zn}$ (C)
- In the above reaction, compounds (A), (B) and (C) are:
- (a) benzene, nitrobenzene and aniline
(b) benzene, dinitrobenzene and m-nitroaniline
(c) toluene, m-nitrobenzene and m-toluidine
(d) benzene, nitrobenzene and hydrabenzene
57. (S) Which of the following is strongly acidic?
- (a) Phenol (b) m-Nitrophenol
(c) m-Bromophenol (d) m-Cresol

58. (S) Salol is prepared from:

- (a) salicylic acid and methyl alcohol
- (b) salicylic acid and phenol
- (c) both (a) and (b)
- (d) none of the above



The product formed in the reaction is:

- (a) benzoquinone
- (b) diphenyl ether
- (c) benzene sulphuric acid
- (d) hydroquinone

60. (S) Phenol on standing in air develops a red colour due to formation of:

- (a) cyclohexane
- (b) resorcinol
- (c) phenoquinone
- (d) quinol

61. (S) Phenol is more acidic than alcohol because:

- (a) phenol is more soluble in polar solvents
- (b) alcohol does not lose hydrogen atom
- (c) phenoxide ion is stabilised by resonance
- (d) phenoxide ion does not exhibit resonance

62. (S) Phenol when condensed with phthalic anhydride in presence of conc. H_2SO_4 , yields:

- (a) methyl orange
- (b) phenolphthalein
- (c) phenol red
- (d) salicylic acid

63. (S) Phenol is heated with CCl_4 and alk. KOH to form salicylic acid. The reaction is known as:

- (a) Friedel-Crafts reaction
- (b) Rosenmund reaction
- (c) Reimer-Tiemann reaction
- (d) Perkin's reaction

64. (S) Phenol reacts with bromine in carbon disulphide at low temperature to give:

- (a) m-bromophenol
- (b) o- and p-bromophenols
- (c) p-bromophenol
- (d) 2,4,6-tribromophenol

65. (S) Phenol is also called:

- (a) salicylic acid
- (b) benzyl alcohol
- (c) carbolic acid
- (d) salol

66. (S) Phenols do not react with:

- (a) sodium bicarbonate
- (b) sodium hydroxide
- (c) potassium hydroxide
- (d) ferric chloride

67. (S) Phenols react with ferric chloride sodium to give:

- (a) blue colouration
- (b) purple colouration
- (c) green colouration
- (d) all of these

68. (S) Phenol is less acidic than:

- (a) p-nitrophenol
- (b) ethanol
- (c) cresol
- (d) benzyl alcohol

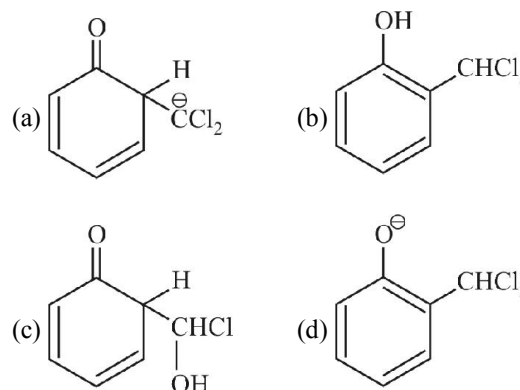
69. (S) Sodium phenoxide when heated with CO_2 under pressure at $125^\circ C$ gives:

- (a) sodium benzoate
- (b) salol
- (c) sodium salicylate
- (d) salicylaldehyde

70. (M) Salicylic acid is prepared from phenol by:

- (a) Reimer-Tiemann reaction
- (b) Kolbe's electrolytic reaction
- (c) Kolbe's reaction
- (d) none of the above

71. (M) When phenol is treated with $CHCl_3$ and NaOH followed by acidification, salicylaldehyde is obtained. Which of the following species are involved in the above mentioned reaction as intermediates?



72. (A) **Assertion:** Phenol is more reactive than benzene towards electrophilic substitution.

Reason: In the case of phenol, the intermediate carbocation is more resonance stabilized.

- (a) A (b) B
(c) C (d) D

73. (A) **Assertion:** o-Phenol sulphonic acid on heating at 100°C changes to p-phenol sulphonic acid

Reason: Sulphonation of phenol is a reversible process.

- (a) A (b) B
(c) C (d) D

74. (A) **Assertion:** Friedal-Crafts reaction between benzene and acetic anhydride in presence of anhydrous AlCl_3 yields acetophenone and not polysubstitution product.

Reason: Acetophenone formed poisons the catalyst, preventing further reaction.

- (a) A (b) B
(c) C (d) D

75. (A) **Assertion** : Phenol undergo Kolbe reaction, ethanol does not.

Reason : Phenoxide ion is more basic than ethoxide ion.

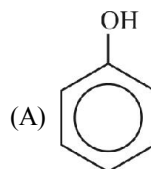
- (a) A (b) B
(c) C (d) D

Mixed Concepts

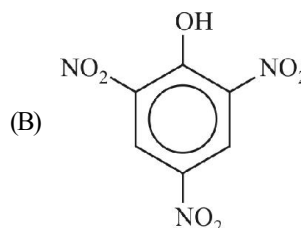
76. (X) Match the following :

Column-I
(Compound)

Column-II
(Possible reaction)



(P) Reaction with NaOH



(Q) Reaction with NaHCO_3

(C) CH_3COOH

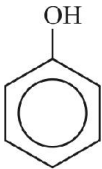
(R) Reaction with PCl_5

(D) $\text{C}_2\text{H}_5\text{OH}$

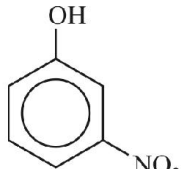
(S) Reaction with NH_3

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

Single Answer Questions

1. Ethyl alcohol is heated with conc. H_2SO_4 . The product formed is (1980)
(a) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$ (b) C_2H_2
(c) C_2H_4 (d) C_2H_6
2. Which of the following is soluble in water. (1980)
(a) CS_2 (b) $\text{C}_2\text{H}_5\text{OH}$
(c) CCl_4 (d) CHCl_3
3. The compound which reacts fastest with Lucas reagent at room temperature is (1981)
(a) butan-2-ol (b) butan-1-ol
(c) 2-methyl propan-1-ol (d) 2-methyl propan-2-ol
4. Diethyl ether on heating with conc. HI gives two moles of (1983)
(a) ethanol (b) iodoform
(c) ethyl iodide (d) methyl iodide
5. When phenol is treated with excess bromine water, it gives (1984)
(a) m-bromophenol (b) o- and p-bromophenol
(c) 2, 4-dibromophenol (d) 2, 4, 6-tribromophenol
6. An industrial method of preparation of methanol is (1984)
(a) catalytic reduction of carbon monoxide in presence of $\text{ZnO}-\text{Cr}_2\text{O}_3$
(b) by reacting methane with steam at 900°C with nickel catalyst
(c) by reducing formaldehyde with LiAlH_4
(d) by reacting formaldehyde with aqueous sodium hydroxide solution
7. HBr reacts fastest with (1986)
(a) 2-methyl propan-2-ol (b) propan-1-ol
(c) propan-2-ol (d) 2-methyl propan-1-ol
8. Hydrogen bonding is maximum in (1987)
(a) ethanol (b) diethyl ether
(c) ethyl chloride (d) triethyl amine
9. In $\text{CH}_3\text{CH}_2\text{OH}$, the bond that undergoes heterolytic cleavage most readily is (1988)
(a) $\text{C}-\text{C}$ (b) $\text{C}-\text{O}$
(c) $\text{C}-\text{H}$ (d) $\text{O}-\text{H}$
10. The products of combustion of an aliphatic thiol (RSH) at 298 K are (1992)
(a) $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\text{g})$ and $\text{SO}_2(\text{g})$
(b) $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\ell)$ and $\text{SO}_2(\text{g})$
(c) $\text{CO}_2(\ell)$, $\text{H}_2\text{O}(\ell)$ and $\text{SO}_2(\text{g})$
(d) $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\ell)$ and $\text{SO}_2(\ell)$
11. In the following compounds (1996)
- 
(I)


(II)


(III)


(IV)
- The order of acidity is
(a) $\text{III} > \text{IV} > \text{I} > \text{II}$ (b) $\text{I} > \text{IV} > \text{III} > \text{II}$
(c) $\text{II} > \text{I} > \text{III} > \text{IV}$ (d) $\text{IV} > \text{III} > \text{I} > \text{II}$
12. $(\text{CH}_3)_3\text{CMgCl}$ on reaction with D_2O produces (1997)
(a) $(\text{CH}_3)_3\text{CD}$ (b) $(\text{CH}_3)_3\text{COD}$
(c) $(\text{CD})_3\text{CD}$ (d) $(\text{CD})_3\text{COD}$

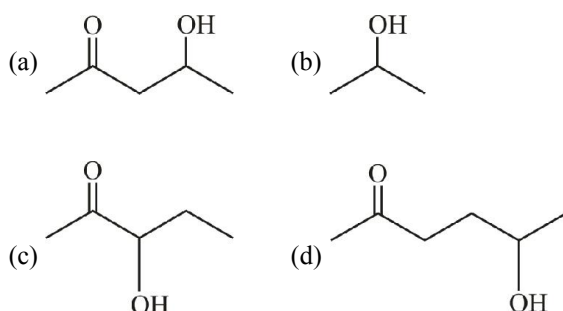
13. Benzene diazonium chloride on reaction with phenol in weakly basic medium gives (1998)

- (a) diphenyl ether (b) p-hydroxy azobenzene
(c) chlorobenzene (d) benzene

14. Among the following compounds, the strongest acid is (1998)

- (a) $\text{HC}\equiv\text{CH}$ (b) C_6H_6
(c) C_2H_6 (d) CH_3OH

15. Which one of the following will most readily be dehydrated in acidic condition? (2000)



16. 1-propanol and 2-propanol can be best distinguished by (2001)

- (a) oxidation with alkaline KMnO_4 followed by reaction with Fehling solution
(b) oxidation with acidic dichromate followed by reaction with Fehling solution
(c) oxidation by heating with copper followed by reaction with Fehling solution
(d) oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution

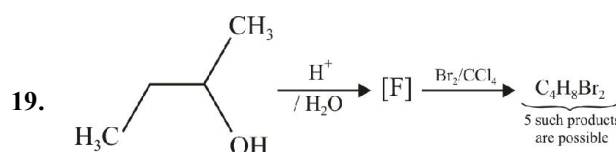
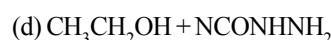
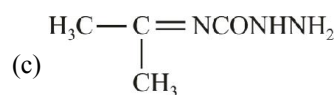
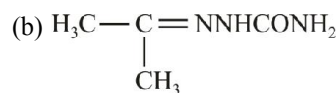
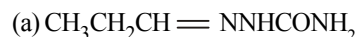
17. The compound that will react most readily with NaOH to form methanol is (2001)

- (a) $(\text{CH}_3)_4\text{N}^+\text{I}^-$ (b) CH_3OCH_3
(c) $(\text{CH}_3)_3\text{S}^+\text{I}^-$ (d) $(\text{CH}_3)_3\text{CI}$

18. Compound 'A' (molecular formula $\text{C}_3\text{H}_8\text{O}$) is treated with acidified potassium dichromate to form a product 'B' (molecular formula $\text{C}_3\text{H}_6\text{O}$) 'B' forms a shining silver mirror on warming with ammoniacal silver nitrate. 'B' when treated

with an aqueous solution of $\text{H}_2\text{NCONHNH}_2$ and sodium acetate gives a product 'C'. Identify the structure of 'C'.

(2002)



How many structures of F is possible? (2003)

- (a) 2 (b) 5
(c) 6 (d) 3

20. When phenyl magnesium bromide reacts with tert. butanol, which of the following is formed? (2005)

- (a) Tert. butyl methyl ether
(b) Benzene
(c) Tert. butyl benzene (d) Phenol

21. The best method to prepare cyclohexene from cyclohexanol is by using

- (a) conc. $\text{HCl} + \text{ZnCl}_2$ (b) conc. H_3PO_4
(c) HBr (d) conc. HCl

22. (I) 1, 2-dihydroxy benzene (2006)

(II) 1, 3-dihydroxy benzene

(III) 1, 4-dihydroxy benzene

(IV) Hydroxy benzene

The increasing order of boiling points of above mentioned alcohols is

- (a) $\text{I} < \text{II} < \text{III} < \text{IV}$ (b) $\text{I} < \text{II} < \text{IV} < \text{III}$
(c) $\text{IV} < \text{I} < \text{II} < \text{III}$ (d) $\text{IV} < \text{II} < \text{I} < \text{III}$

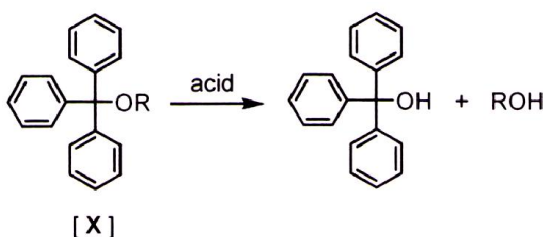
23. $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 + \text{KOH} \rightarrow$ Nitrogen containing compound + $\text{KCl} + \text{H}_2\text{O}$. Nitrogen containing compound is

(2006)

- (a) $\text{CH}_3-\text{C}\equiv\text{N}$ (b) $\text{CH}_3-\text{NH}-\text{CH}_3$
(c) $\text{CH}_3-\text{N}^{\ominus}\equiv\text{C}^{\oplus}$ (d) $\text{CH}_3\text{N}^{\oplus}\equiv\text{C}^{\ominus}$

24. The acidic hydrolysis of ether (X) shown below is fastest when

(2014)



- (a) one phenyl group is replaced by a methyl group.
(b) one phenyl group is replaced by a para-methoxyphenyl group.
(c) two phenyl groups are replaced by two para-methoxyphenyl groups.
(d) no structural change is made of X.

Assertion Reason

- (A) If both ASSERTION and REASON are true and reason is the correct explanation of the assertion.
(B) If both ASSERTION and REASON are true but reason is not the correct explanation of the assertion.
(C) If ASSERTION is true but REASON is false.
(D) If both ASSERTION and REASON are false.
(E) If ASSERTION is false but REASON is true.
25. **Assertion :** Solubility of n-alcohol in water decreases with increase in molecular weight.

Reason : The reactive proportion of the hydrocarbon part in alcohols increases with increasing molecular weight which permits enhanced hydrogen bonding with water. (1988)

26. **Assertion :** p-nitrophenol is a stronger acid than o-nitrophenol.

Reason : Intramolecular hydrogen bonding make the o-isomer weaker than the p-isomer. (1989)

27. **Assertion :** Bromobenzene, upon reaction with Br_2/Fe gives 1, 4-dibromobenzene as the major product.

Reason : In bromobenzene, the inductive effect of the bromo group is more dominant than the mesomeric effect in directing the incoming electrophile. (2008)

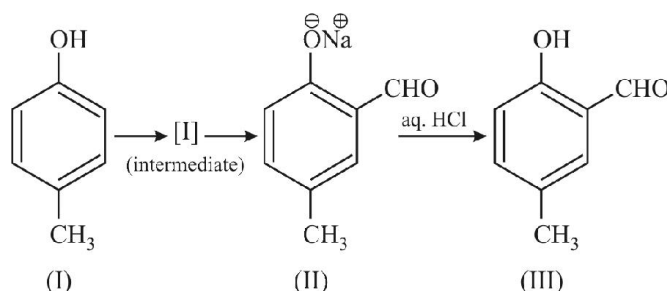
Comprehension

Using the following Passage, solve Q. 28 to Q. 30

Passage – 1

Reimer-Tiemann reaction introduces an aldehyde group, on to the aromatic ring of phenol, ortho to the hydroxyl group. This reaction involves electrophilic aromatic substitution. This is a general method for the synthesis of substituted salicylaldehydes as depicted below :

(2007)



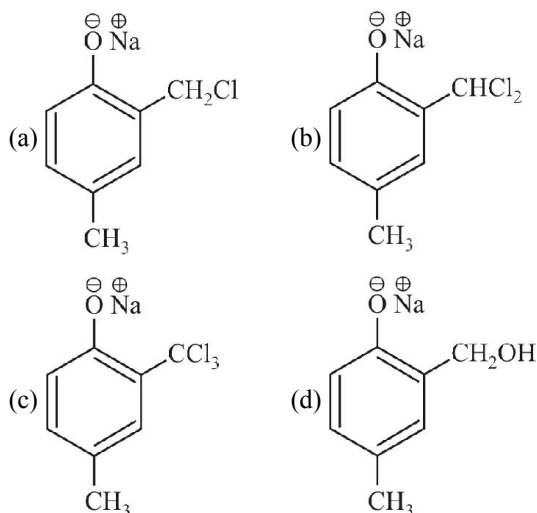
28. Which one of the following reagents is used in the above reaction ?

- (a) aq. $\text{NaOH} + \text{CH}_3\text{Cl}$
(b) aq. $\text{NaOH} + \text{CH}_2\text{Cl}_2$
(c) aq. $\text{NaOH} + \text{CHCl}_3$
(d) aq. $\text{NaOH} + \text{CCl}_4$

29. The electrophile in this reaction is

- (a) : CHCl (b) $^{\oplus}\text{CHCl}_2$
(c) : CCl_2 (d) $\bullet\text{CCl}_3$

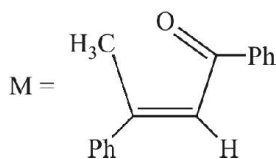
30. The structure of the intermediate I is



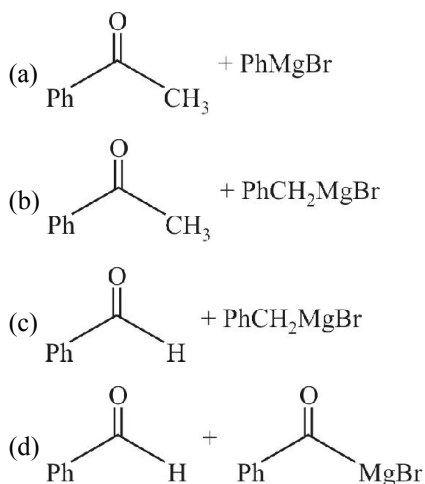
Using the following Passage, solve Q. 31 to Q. 33

Passage – 2

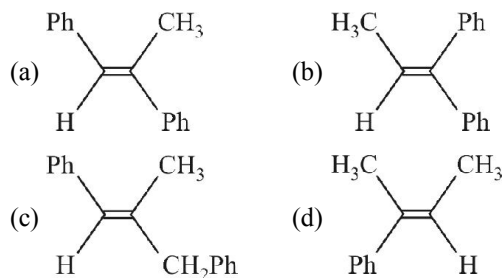
A tertiary alcohol H upon acid catalysed dehydration gives a product I. Ozonolysis of I leads to compounds J and K. Compound J upon reaction with KOH gives benzyl alcohol and a compound L, where K on reaction with KOH gives only M.



31. Compound H is formed by the reaction of



32. The structure of compound I is

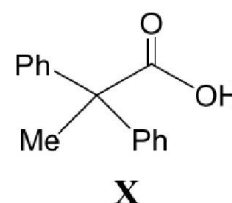


33. The structures of compounds J, K and L respectively, are

- (a) PhCOCH₃, PhCH₂COCH₃ and PhCH₂COO⁻K⁺
 (b) PhCHO, PhCH₂CHO and PhCH₂COO⁻K⁺
 (c) PhCOCH₃, PhCH₂CHO and CH₃COO⁻K⁺
 (d) PhCHO, PhCOCH₃ and PhCOO⁻K⁺

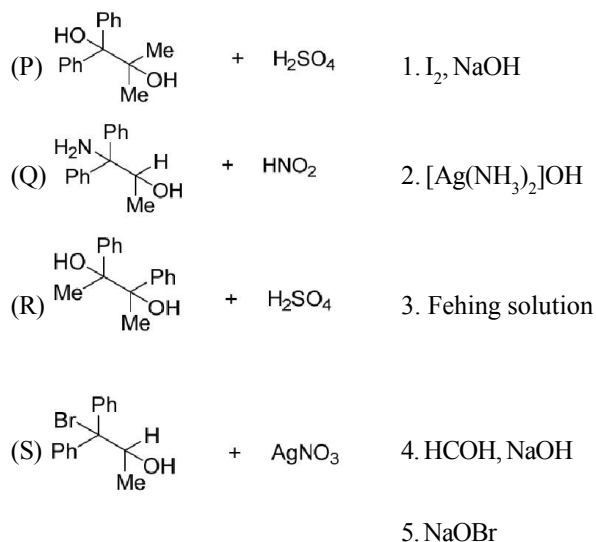
34. The desired product X can be prepared by reacting the major product of the reaction in List-I with one or more appropriate reagents in List-II

(given, order of migratory aptitude : aryl > alkyl > hydrogen)



List-I

List-II



The correct option is

- (a) P - 1; Q - 2,3; R - 1, 4; S - 2, 4
 (b) P - 1, 5; Q - 3,4; R - 4, 5; S - 3
 (c) P - 1,5; Q - 3,4; R - 5; S - 2, 4
 (d) P - 1, 5; Q - 2, 3; R - 1,5; S - 2, 3

(2018)

Match the Column

35. Match the following choosing one item from column 'X' and the appropriate item from column Y. (1983)

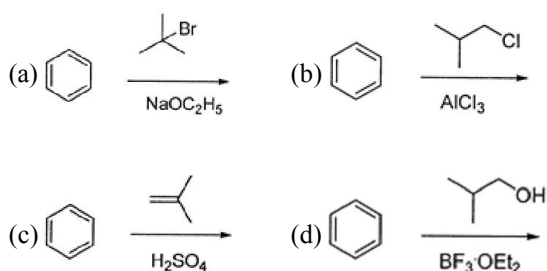
X

Y

- | | |
|----------------------------------|------------------|
| (A) Williamson's synthesis | (P) tert-alcohol |
| (B) Neutral FeCl_3 test | (Q) Ether |
| (C) Lucas test | (R) Phenol |

Multiple Answer Questions

36. Among the following reactions(s) which gives(give) tert-butyl benzene as the major product is(are) (2016)



37. The correct combination of names for isomeric alcohols with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ is/are (2014)

- (a) tert-butanol and 2-methylpropan-2-ol
(b) tert-butanol and 1, 1-dimethylethan-1-ol
(c) n-butanol and butan-1-ol
(d) isobutyl alcohol and 2-methylpropan-1-ol

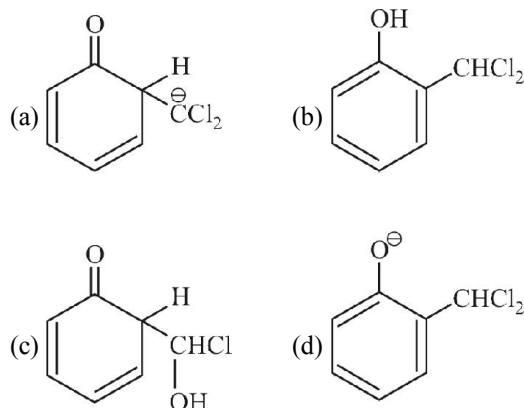
38. Phenol is less acidic than (1986)

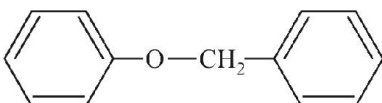
- (a) acetic acid (b) p-methoxyphenol
(c) p-nitrophenol (d) ethanol

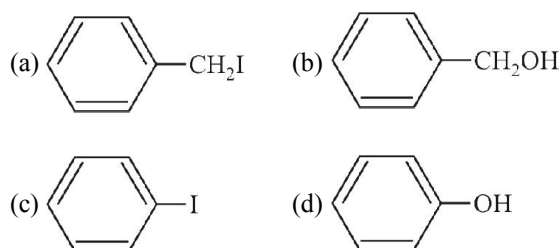
39. The products of reaction of alcoholic silver nitrate with ethyl bromide are (1991)

- (a) ethane (b) ethene
(c) nitroethane (d) ethyl nitrite

40. When phenol is reacted with CHCl_3 and NaOH followed by acidification, salicylaldehyde is obtained. Which of the following species are involved in the above mentioned reaction as intermediates? (1995)



41. The ether  when treated with HI produces (1999)



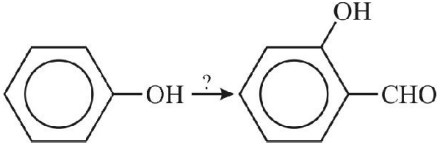
Fill in the Blanks

42. Ethanol vapour is passed over heated copper and the product is treated with aqueous NaOH. the final product is (1983)
43. A diol has two hydroxyl groups on carbon atoms. (1986)
44. Aliphatic ethers are purified by shaking with a solution of ferrous salt to remove which are formed on prolonged standing in contact with water. (1992)
45. Glycerine contains one hydroxyl group. (1997)

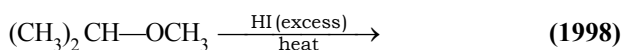
True/False

46. The yield of a ketone when a secondary alcohol is oxidised is more than the yield of aldehyde when a primary alcohol is oxidised. (1983)
47. Sodium ethoxide is prepared by reacting ethanol with aqueous sodium hydroxide. (1985)

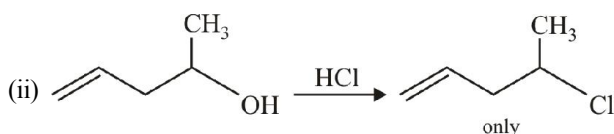
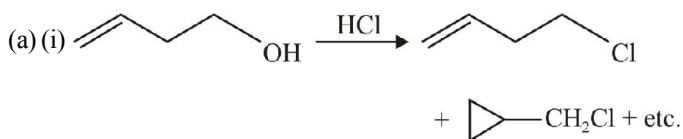
Subjective Questions

48. An organic liquid 'A' containing C, H and O with boiling point 78°C , possessing a rather pleasant odour, on heating with concentrated sulphuric acid gives a gaseous product 'B' with the empirical formula, CH_2 . 'B' decolourises bromine water as well as alkaline permanganate solution and takes up one mole of H_2 (per mole of B) in presence of finely divided nickel at high temperature. Identify the substances A and B. (1979)
49. A compound 'X' containing C, H and O is unreactive towards sodium. It does not add with bromine. It also does not react with Schiff's reagent. On refluxing with an excess of HI, 'X' yields only one organic product 'Y'. 'Y' on hydrolysis yields a new compound 'Z' which can be converted into 'Y' by reaction with red phosphorus and iodine. The compound 'Z' on oxidation with potassium permanganate gives a carboxylic acid. The equivalent weight of acid is 60. What are the compounds 'X', 'Y' and 'Z'? Write chemical equations leading to the conversion of 'X' to 'Y'. (1981)
50. State the conditions under which the following preparations are carried out. Give necessary equations which need not be balanced.
- Ethanol from acetylene
 - Lead tetrathyl from sodium-lead alloy
 - Methyl chloride from aluminium carbide (1983)
51. An alcohol A, when heated with conc. H_2SO_4 gives an alkene B. When B is bubbled through bromine water and the product obtained is dehydrohalogenated with excess of sodamide, a new compound C is obtained. The compound C gives D when treated with warm dilute H_2SO_4 in presence of HgSO_4 . D can also be obtained either by oxidising A with KMnO_4 or from acetic acid through its calcium salt. Identify A, B, C and D. (1983)
52. Suggest a reason for the large difference between the boiling points of butanol and butanal, although they have almost the same solubility in water. (1985)
53. A compound of molecular formula $\text{C}_7\text{H}_8\text{O}$ is insoluble in water and dilute sodium bicarbonate but dissolves in dilute aqueous sodium hydroxide and gives a characteristic colour with aqueous FeCl_3 . On treatment with bromine water, it readily gives a precipitate of $\text{C}_7\text{H}_5\text{OBr}_3$. Write down the structure of the compound. (1985)
54. Give a chemical test to distinguish between methanol and ethanol. (1985)
55. Write down the main product of the following reaction :
Ethanol $\xrightarrow{\text{I}_2/\text{NaOH}}$ (1985)
56. How may be the following transformation be carried out (in not more than six steps) ? (1986)
"Ethyl alcohol to vinyl acetate."
57. Complete the following with appropriate structures :
- 
58. Explain the following in one or two sentences only :
"Phenol is an acid, but it does not react with sodium bicarbonate." (1987)
59. Arrange the following in increasing order of boiling point :
n-butane, n-butanol, n-butyl chloride, iso-butane. (1988)
60. Indicate steps which would convert : (1989)
(i) phenol to acetophenone
(ii) acetic acid to tert-butyl alcohol
61. How will you bring about the following conversion ?
"Ethanal to 2-hydroxy-3-butenic acid." (1990)
62. Give reason in one or two sentences for the following :
"Iodoform is obtained by the reaction of acetone with hypoiodite but not with iodide." (1991)
63. Compound X (molecular formula, $\text{C}_5\text{H}_8\text{O}$) does not react appreciably with Lucas reagent at room temperature but gives a precipitate with ammoniacal silver nitrate. With excess of MeMgBr , 0.42 g of X gives 224 mL of CH_4 at STP. Treatment of X with H_2 in presence of Pt catalyst followed by boiling with excess HI, gives n-pentane. Suggest structure for X and write the equation involved. (1992)

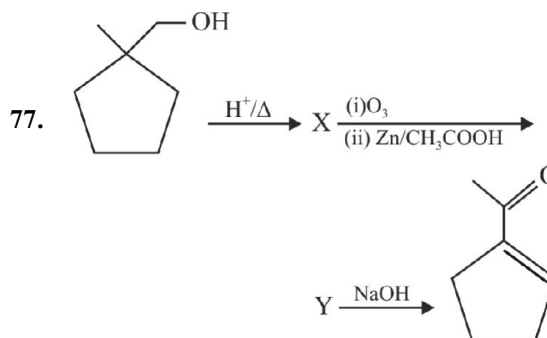
64. When t-butanol and n-butanol are separately treated with a few drops of dilute KMnO_4 in one case only, the purple colour disappears and a brown precipitate is formed. Which of the two alcohols gives the above reaction and what is the brown precipitate? (1994)
65. 3, 3-dimethylbutan-2-ol loses a molecule of water in the presence of concentrated sulphuric acid to give tetramethylethylene as a major product. Suggest a suitable mechanism. (1996)
66. A compound D ($\text{C}_8\text{H}_{10}\text{O}$) upon treatment with alkaline solution of iodine gives a yellow precipitate. The filtrate on acidification gives a white solid E ($\text{C}_7\text{H}_6\text{O}_2$). Write the structures of D, E and explain the formation of E. (1996)
67. 2, 2-dimethyloxirane can be cleaved by acid (H^+). Write mechanism. (1997)
68. An optically active alcohol A ($\text{C}_6\text{H}_{10}\text{O}$) absorbs two moles of hydrogen per mole of A upon catalytic hydrogenation and gives a product B. The compound B is resistant to oxidation by CrO_3 and does not show any optical activity. Deduce the structure of A and B. (1996)
69. 2, 2-dimethyloxirane can be cleaved by acid (H^+). Write mechanism. (1997)
70. Which of the following is the correct method for synthesizing methyl-t-butyl ether and why? (1997)
- (i) $(\text{CH}_3)_3\text{CBr} + \text{NaOMe} \longrightarrow$
- (ii) $\text{CH}_3\text{Br} + \text{NaO-t-Bu} \longrightarrow$
71. Although phenoxide ion has more number of resonating structures than benzoate ion, benzoic acid is a stronger acid than phenol. Why? (1997)
72. Write the structures of the products :



73. Explain briefly the formation of products giving the structures of the intermediates.

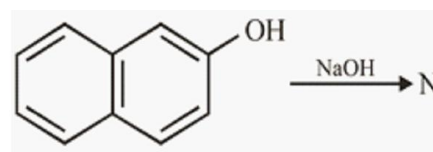


74. Give reasons for the following in one or two sentences.
- “Acid catalysed dehydration of t-butanol is faster than that of n-butanol.” (1998)
75. Cyclobutylbromide on treatment with magnesium in dry ether forms an organometallic compound (A). The organometallic reacts with ethanal to give an alcohol (B) after mild acidification. Prolonged treatment of alcohol (B) with an equivalent amount of HBr gives 1-bromo-1-methylcyclopentane (C). Write the structures of (A), (B) and explain how (C) is obtained from (B). (2001)
76. An organic compound ‘P’ having the molecular formula $\text{C}_5\text{H}_{10}\text{O}$ when treated with dil H_2SO_4 gives two compounds, Q & R both gives positive iodoform test. The reaction of $\text{C}_5\text{H}_{10}\text{O}$ with dil H_2SO_4 gives reaction 10^{15} times faster than ethylene. Identify organic compound of Q and R. Give the reason for the extra stability of P. (2004)



Identify X and Y. (2005)

78. The number of resonance structures for N is (2015)



ANSWER KEY

EXERCISE - 1 (Basic Objective Questions)

1. (b)	2. (a)	3. (c)	4. (d)	5. (d)	6. (c)	7. (c)	8. (c)	9. (d)	10. (c)
11. (c)	12. (a)	13. (d)	14. (a)	15. (d)	16. (a)	17. (b)	18. (c)	19. (b)	20. (d)
21. (a)	22. (a)	23. (d)	24. (a)	25. (a)	26. (b)	27. (b)	28. (a)	29. (b)	30. (d)
31. (c)	32. (c)	33. (b)	34. (a)	35. (d)	36. (b)	37. (a)	38. (d)	39. (d)	40. (b)
41. (b)	42. (b)	43. (d)	44. (a)	45. (a)	46. (d)	47. (d)	48. (a)	49. (a, d)	50. (b)
51. (b)	52. (a)	53. (b)	54. (a)	55. (c)	56. (c)	57. (c)	58. (d)	59. (b)	60. (d)
61. (a)	62. (b)	63. (c)	64. (a)	65. (a)	66. (b)	67. (a)	68. (a)	69. (b)	70. (c)

EXERCISE - 2 (Previous Year JEE MAINS Questions)

1. (a)	2. (d)	3. (c)	4. (b)	5. (c)	6. (a)	7. (b)	8. (b)	9. (c)	10. (b)
11. (d)	12. (a)	13. (b)	14. (c)	15. (d)	16. (d)	17. (d)	18. (c)	19. (b)	20. (a)
21. (b)									

JEE Mains Online

1. (c)	2. (b)	3. (b)	4. (d)	5. (b)	6. (d)	7. (d)	8. (c)	9. (a)	10. (a)
11. (b)	12. (d)	13. (b)	14. (c)						

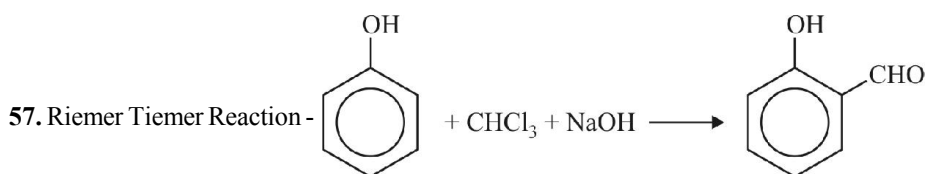
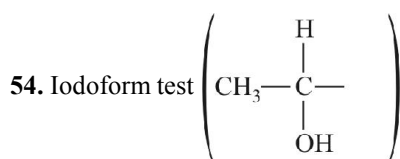
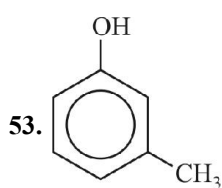
EXERCISE - 3 (Advanced Objective Questions)

1. (c)	2. (d)	3. (a)	4. (c)	5. (c)	6. (b)	7. (a)	8. (c)	9. (c)	10. (d)
11. (c)	12. (a)	13. (c)	14. (c)	15. (a)	16. (d)	17. (a,b,c)	18. (a,b)	19. (a,b)	20. (a,b,c,d)
21. (d)	22. (a)	23. (c)	24. (c)	25. (c)	26. (d)	27. (c)	28. (c)	29. (a)	30. (d)
31. (a)	32. (a)	33. (c)	34. (a)	35. (c)	36. (a,b)	37. (c)	38. (a)	39. (b)	40. (a)
41. (c)	42. (b)	43. (a)	44. (c)	45. (c)	46. (b,d)	47. (a,d)	48. (b,c,d)	49. (a,c,d)	50. (b,c)
51. (d)	52. (a)	53. (b)	54. (c)	55. (c)	56. (d)	57. (b)	58. (b)	59. (d)	60. (c)
61. (c)	62. (b)	63. (c)	64. (b)	65. (c)	66. (a)	67. (d)	68. (a)	69. (c)	70. (a,c)
71. (a,d)	72. (a)	73. (b)	74. (c)	75. (c)	76. A – P, R, S ; B – P, Q, R, S ; C – P, Q, R, S ; D – P, R, S				

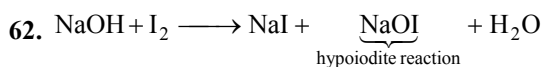
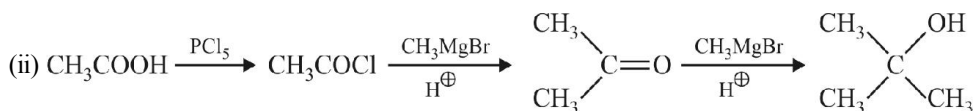
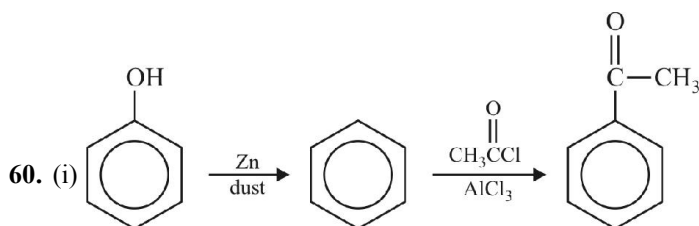
EXERCISE - 4 (Previous Year JEE Advanced Questions)

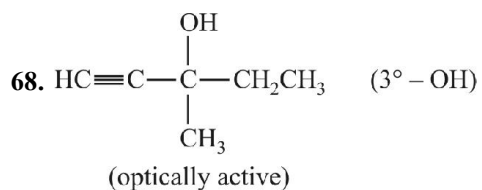
1. (c) 2. (b) 3. (d) 4. (c) 5. (d) 6. (a) 7. (a) 8. (a) 9. (d) 10. (b)
 11. (d) 12. (a) 13. (b) 14. (d) 15. (a) 16. (c) 17. (a) 18. (a) 19. (d) 20. (b)
 21. (b) 22. (c) 23. (d) 24. (c) 25. (c) 26. (a) 27. (c) 28. (c) 29. (c) 30. (b)
 31. (b) 32. (a) 33. (d) 34. (d) 35. (A) – Q ; (B) – R ; (C) – P 36. (b,c,d) 37. (a,c,d) 38. (a,c)
 39. (c, d) 40. (a,d) 41. (a,d) 42. aldol 43. vicinal; same 44. peroxides 45. secondary
 46. (F) 47. (F) 48. A is ethanol B is an alkene

52. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} \rightarrow$ Intermolecular H_2 bonding. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO} \rightarrow$ No Bonding.

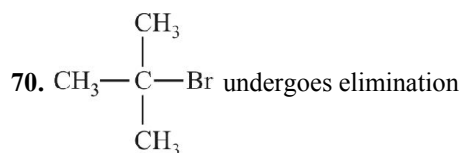


58. Phenol is a weaker acid than NaHCO_3 .





69. Formation of a 3° carbocation.



CH_3Br undergoes $\text{S}_{\text{N}}1$.

71. Benzoate ion has two equivalent resonating structures with negative charge on oxygen atom. In phenoxide, negative charge is delocalized on only one oxygen & three comparatively much less electronegative carbon atoms.
74. 3° carbocation formation is formed in dehydration of t-butanol.

Dream on !!

